

Prevails and Mirages of Large Language Models in Clinical NLP

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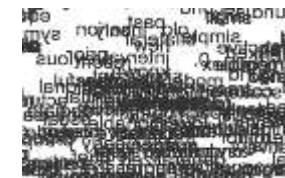
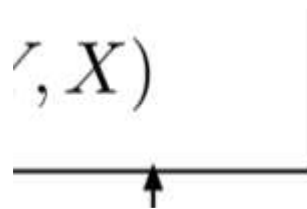
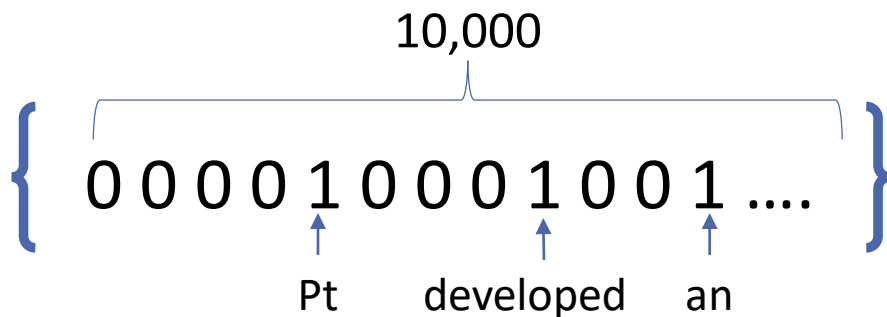
Language model

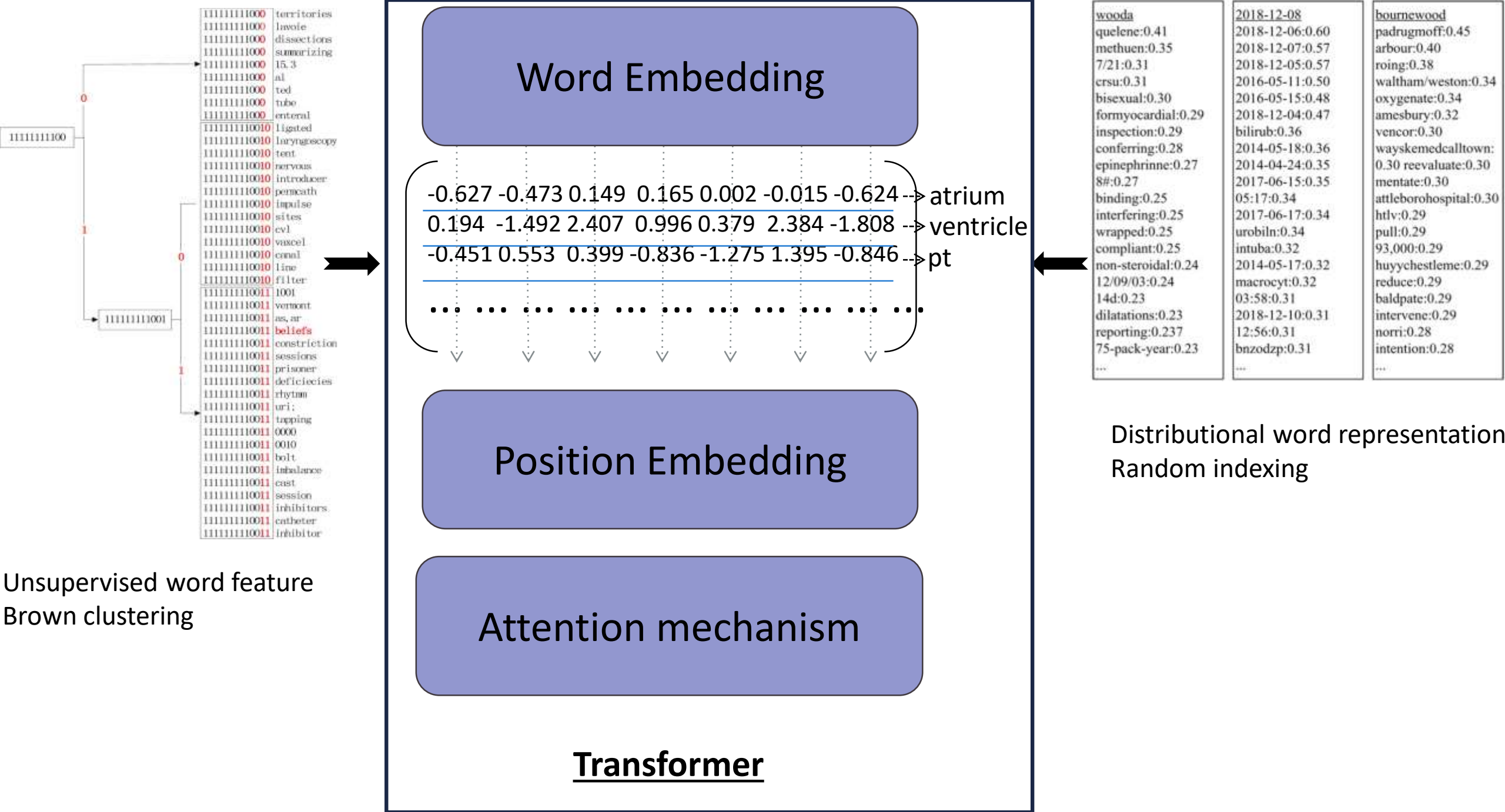
- A statistical distribution to formulate the probability of a sequence or the next word in a sequence
 - $P(W_1, W_2, \dots, W_n)$
- “*A Neural Language Model*”, Bengio et al., 2003
- A self-supervised learning objective to train transformer architecture
- When the model size is very large, e.g., hundreds of billions, interesting machine intelligence emerge:
 - Transfer learning - learn knowledge from one task and apply it to another task
 - Few-shot learning - learn from very few labeled samples
 - Zero-shot learning - learn without labeled samples for the target application
- The architecture is known as “Large Language Model”

Deep learning struggle in NLP



Bag-of-words representation
Vocabulary size: 10,000

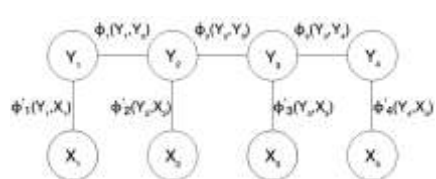




Unified learning architecture

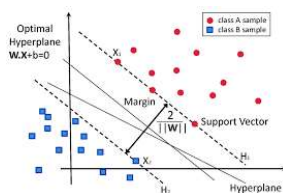
Fully supervised learning

- One model per task
- Output controlled by model parameter



$$P(y|x; \theta)$$

1. Clinical concept extraction
The patient had a moderately dilated aortic ...

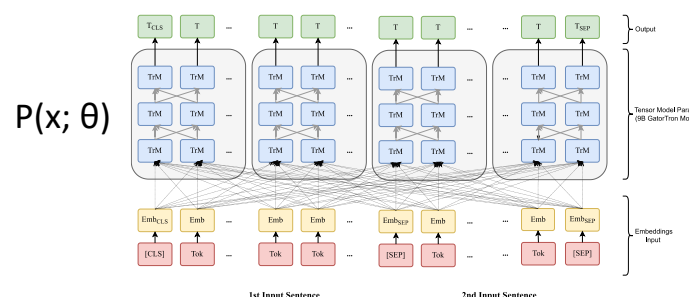


$$P(y|x; \theta)$$

2. Drug-Adverse event relations
Thalidomide discontinued secondary to skin rash ...

Pretrain | fine-tune

- Unified pretrained model
- One output layer per task
- Output controlled by model parameter



Fine-tune

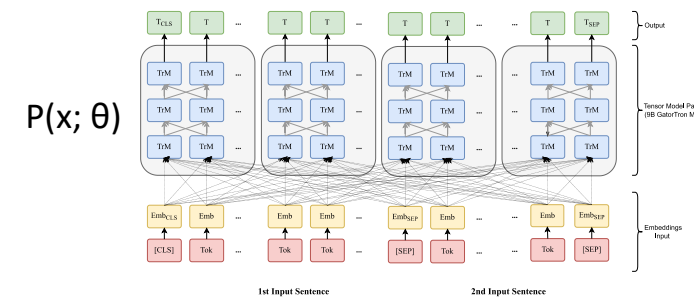
Task specific output layer

$$P(y|x; \theta)$$

1. Clinical concept extraction
The patient had a moderately dilated aortic ...
2. Drug-Adverse event relations
Thalidomide discontinued secondary to skin rash ...
3. Semantic similarity= 0
S1: Patient discharged ambulatory without further questions;
S2: Please contact location at phone number with any questions
4. Natural Language Inference
"Labs were notable for Cr 1.7" ! "Patient has elevated Cr"
5. Question Answering
What lab results shows patient have diabetes?

Pretrain | Prompting

- Unified pretrained model
- Unified output layer
- Output controlled by prompts



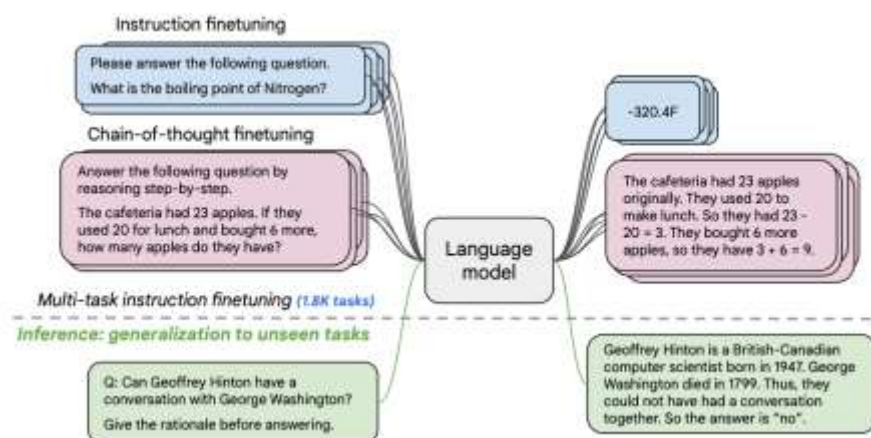
Prompt

$$P(y|f_{\text{prompt}}(x); \theta)$$

1. Clinical concept extraction
The patient had a moderately dilated aortic ...
2. Drug-Adverse event relations
Thalidomide discontinued secondary to skin rash ...
3. Semantic similarity= 0
S1: Patient discharged ambulatory without further questions;
S2: Please contact location at phone number with any questions
4. Natural Language Inference
"Labs were notable for Cr 1.7" ! "Patient has elevated Cr"
5. Question Answering
What lab results shows patient have diabetes?

Generative LLMs

- A general form of AI – artificial general intelligence (AGI)
- Pretrain an LLM through self-supervised learning
- Align the LLM to human instructions through fine-tuning
 - Reinforcement learning with human feedback (RLHF)
 - Reward model
 - Multi-task instruction tuning



<https://arxiv.org/pdf/2210.11416>

Step 1
Collect demonstration data,
and train a supervised policy.

A prompt is
sampled from our
prompt dataset.

Explain the moon
landing to a 6-year-old

A labeler
demonstrates the
desired output
behavior.

Some people went
to the moon...

This data is used
to fine-tune GPT-3
with supervised
learning.

SFT

Step 2
Collect comparison data,
and train a reward model.

A prompt and
several model
outputs are
sampled.

Explain the moon
landing to a 6-year-old:

Explain the moon
landing to a 6-year-old:
Some people went to the moon...
Some people went to the moon...
Some people went to the moon...

A labeler ranks
the outputs from
best to worst.

Some people went to the moon...
Some people went to the moon...
Some people went to the moon...

This data is used
to train our
reward model.

RM

Step 3
Optimize a policy against
the reward model using
reinforcement learning.

A new prompt
is sampled from
the dataset.

Write a story
about frogs

The policy
generates
an output.

PPO

Once upon a time...

The reward model
calculates a
reward for
the output.

RM

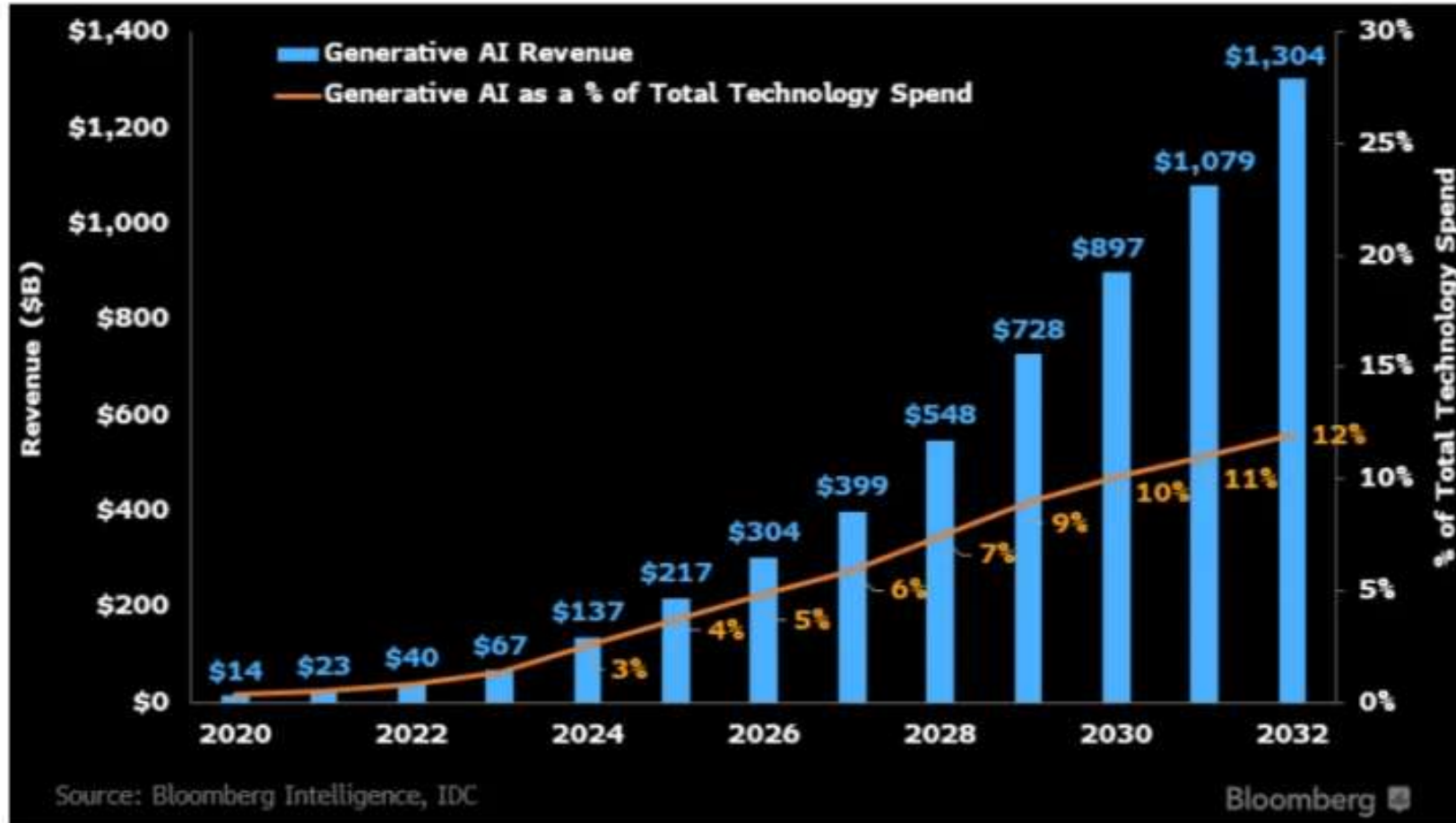
The reward is
used to update
the policy
using PPO.

r_k

<https://arxiv.org/abs/2203.02155>

LLM: Game of Thrones of IT Giants

The Scale-up law



<https://www.bloomberg.com/company/press/generative-ai-to-become-a-1-3-trillion-market-by-2032-research-finds/>



Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond. <https://arxiv.org/abs/2304.13712>

GatorTron models : 345 Million, 3.9 Billion, and 8.9 Billion parameters

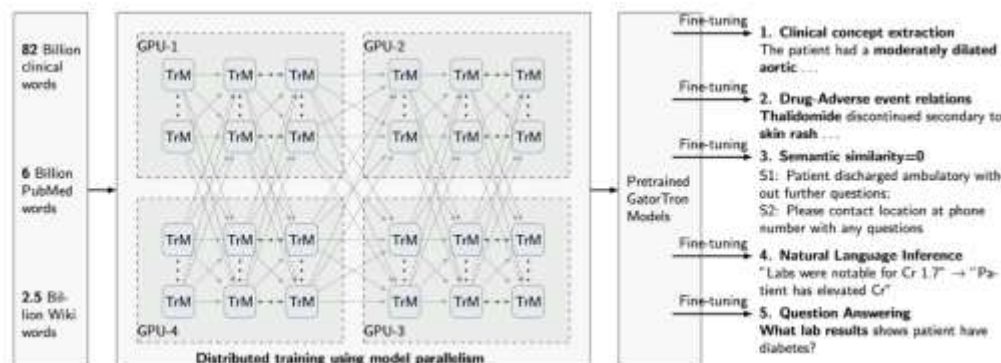
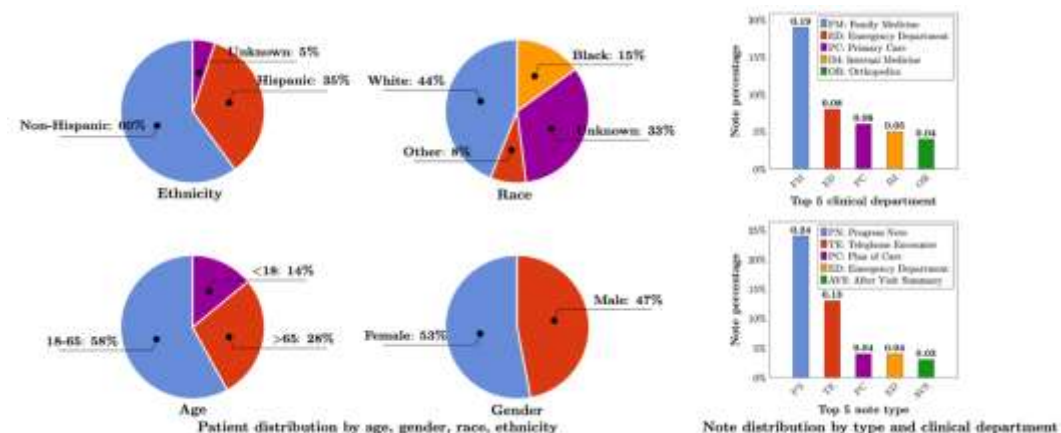
ARTICLE OPEN

A large language model for electronic health records

Xi Yang^{1,2}, Aokun Chen^{1,2}, Nima PourNejatian³, Hoo Chang Shin³, Kaleb E. Smith³, Christopher Parisien³, Colin Compas³, Cheryl Martin³, Anthony B. Costa³, Mona G. Flores^{1,5}, Ying Zhang^{1,6}, Tanja Magoc⁵, Christopher A. Harle^{1,5}, Gloria Lipori^{5,6}, Duane A. Mitchell⁶, William R. Hogan^{1,6}, Elizabeth A. Shenkman^{1,6}, Jiang Bian^{1,2} and Yonghui Wu^{1,2}

There is an increasing interest in developing artificial intelligence (AI) systems to process and interpret electronic health records (EHRs). Natural language processing (NLP) powered by pretrained language models is the key technology for medical AI systems utilizing clinical narratives. However, there are few clinical language models, the largest of which trained in the clinical domain is comparatively small at 110 million parameters (compared with billions of parameters in the general domain). It is not clear how large clinical language models with billions of parameters can help medical AI systems utilize unstructured EHRs. In this study, we develop from scratch a large clinical language model—GatorTron—using >90 billion words of text (including >82 billion words of de-identified clinical text) and systematically evaluate it on five clinical NLP tasks including clinical concept extraction, medical relation extraction, semantic textual similarity, natural language inference (NLI), and medical question answering (MQA). We examine how (1) scaling up the number of parameters and (2) scaling up the size of the training data could benefit these NLP tasks. GatorTron models scale up the clinical language model from 110 million to 8.9 billion parameters and improve five clinical NLP tasks (e.g., 9.6% and 9.5% improvement in accuracy for NLI and MQA), which can be applied to medical AI systems to improve healthcare delivery. The GatorTron models are publicly available at: https://catalog.ngc.nvidia.com/orgs/nvidia/teams/clara/models/gatortron_og.

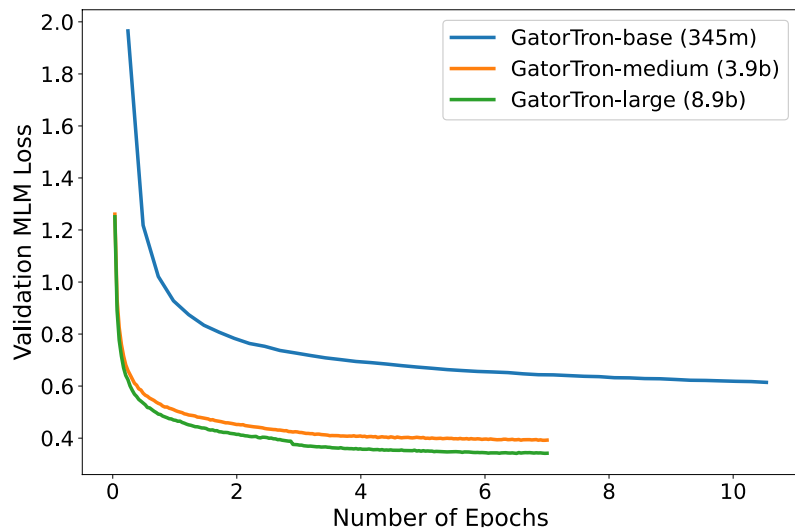
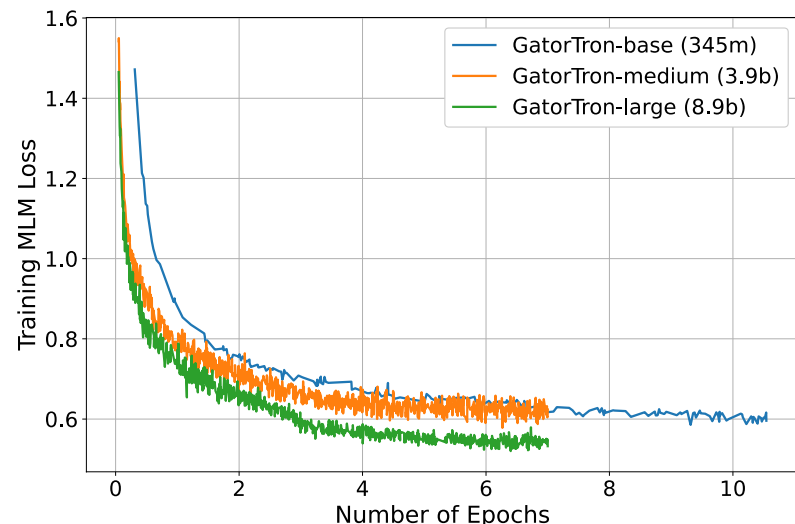
npj Digital Medicine (2022)5:194; <https://doi.org/10.1038/s41746-022-00742-2>



HiperGator-AI with 124 nodes; 992 GPUs; 6 days
Available from: <https://huggingface.co/UFNLP>
Over 1.2 Million downloads

Model	# Layers	# Hidden Size	# Attention Heads	# Parameters
GatorTron-base	24	1024	16	345 million
GatorTron-medium	48	2560	40	3.9 billion
GatorTron-large	56	3584	56	8.9 billion

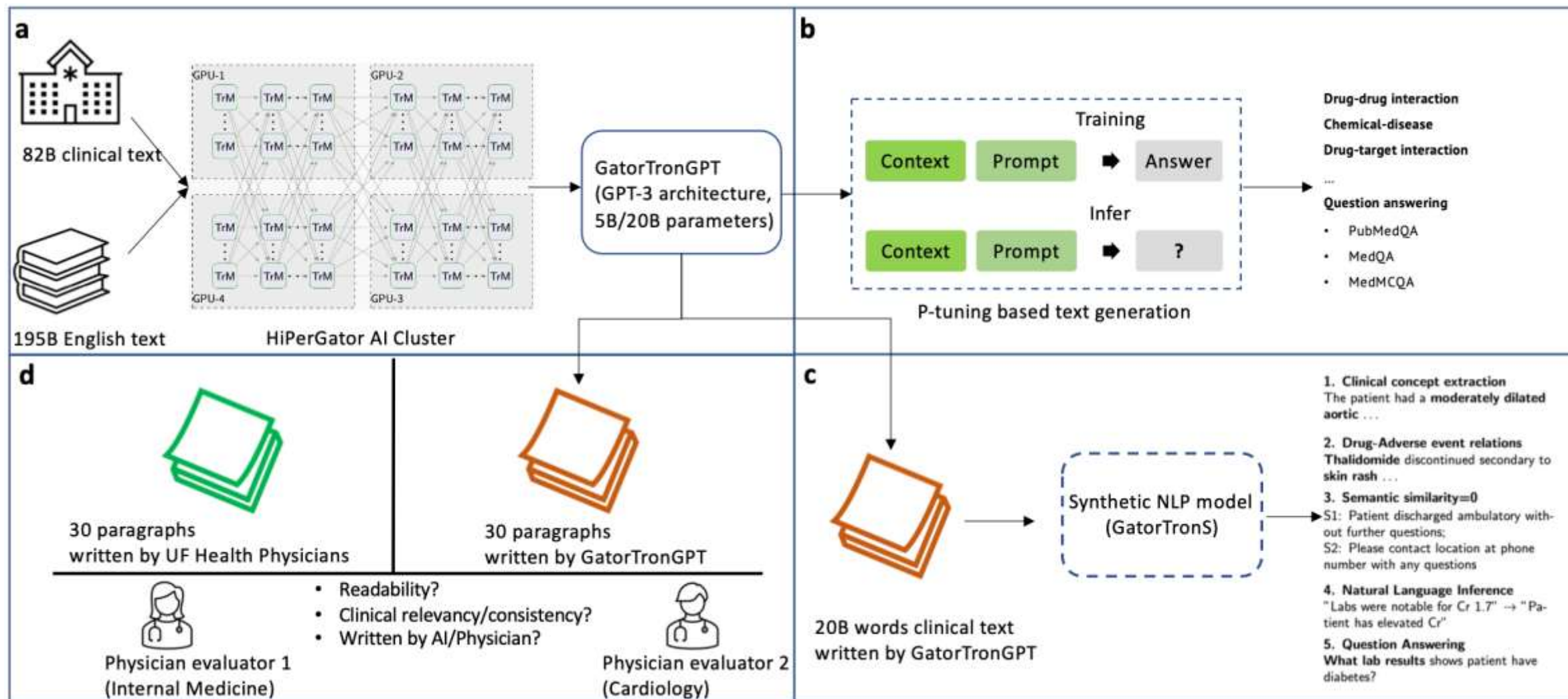
One LLM solves 5 clinical NLP tasks – fine tuning



	Clinical concept extraction									Medical relation extraction		
	2010 i2b2 ³⁹			2012 i2b2 ⁴⁰			2018 n2c2 ⁴¹			2018 n2c2 ⁴¹		
Transformer	Precision	Recall	F1 score	Precision	Recall	F1 score	Precision	Recall	F1 score	Precision	Recall	F1 score
BioBERT	0.8693	0.8653	0.8673	0.7478	0.8037	0.7747	0.8634	0.8921	0.8775	0.9663	0.9451	0.9555
ClinicalBERT	NA	NA	0.8780	NA	NA	0.7890	0.8592	0.8832	0.8710	0.9678	0.9414	0.9544
BioMegatron	0.8614	0.8761	0.8687	0.7591	0.8031	0.7805	0.8707	0.8915	0.8810	0.9711	0.9434	0.9571
GatorTron-base (1/4 data)	0.8682	0.9046	0.8860	0.7514	0.8013	0.7755	0.8772	0.8992	0.8881	0.9724	0.9457	0.9589
GatorTron-base	0.8748	0.9043	0.8893	0.7644	0.8221	0.7922	0.8759	0.9038	0.8896	0.9719	0.9482	0.9599
GatorTron-medium	0.8869	0.9122	0.8994	0.7812	0.8245	0.8022	0.8954	0.9035	0.8994	0.9721	0.9503	0.9611
GatorTron-large	0.8880	0.9116	0.8996	0.7862	0.8333	0.8091	0.8979	0.9021	0.9000	0.9776	0.9482	0.9627

	Semantic textual similarity	Natural language inference	Question answering			
	2019 n2c2 ⁴²	MedNLI ⁴³	emrQA Medication ⁴⁴		emrQA Relation ⁴⁴	
Transformer	Pearson correlation	Accuracy	F1 score	Exact Match	F1 score	Exact Match
BioBERT	0.8744	0.8050	0.6997	0.2475	0.9262	0.8361
ClinicalBERT	0.8787	0.8270	0.6905	0.2406	0.9306	0.8533
BioMegatron	0.8806	0.8390	0.7231	0.2882	0.9405	0.879
GatorTron-base (1/4 data)	0.8675	0.8643	0.7281	0.2952	0.9390	0.8579
GatorTron-base	0.8810	0.8670	0.7181	0.2978	0.9543	0.9029
GatorTron-medium	0.8903	0.8720	0.7354	0.3018	0.9677	0.9243
GatorTron-large	0.8896	0.9020	0.7408	0.3155	0.9719	0.9310

GatorTronGPT - A generative LLM for EHRs

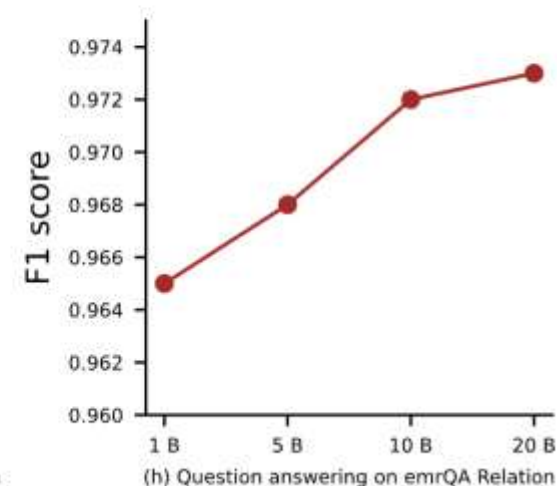
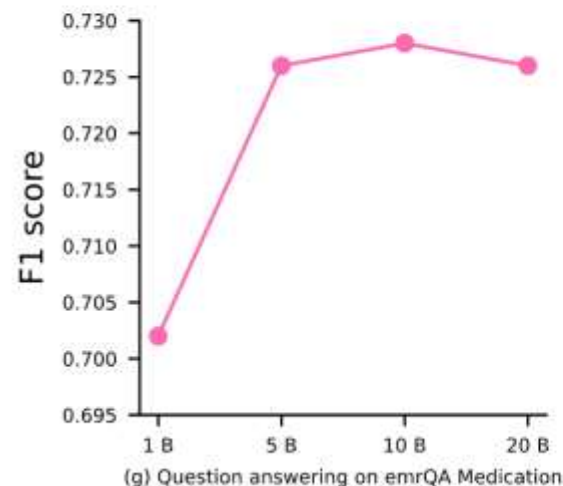
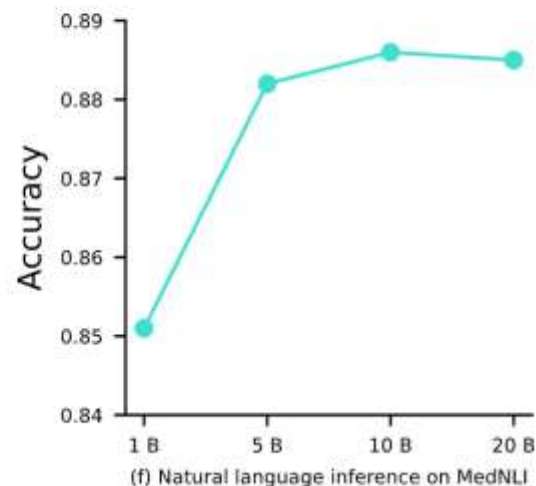
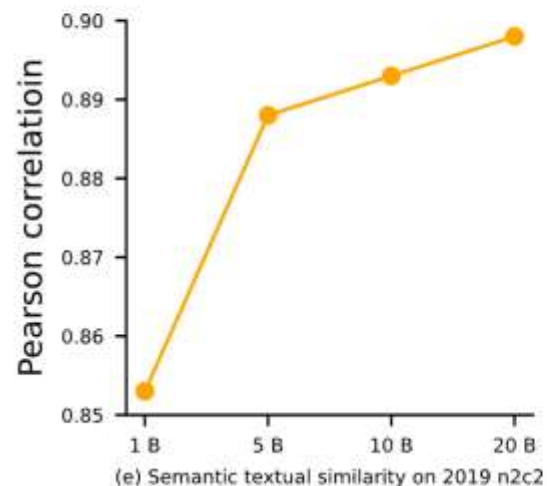
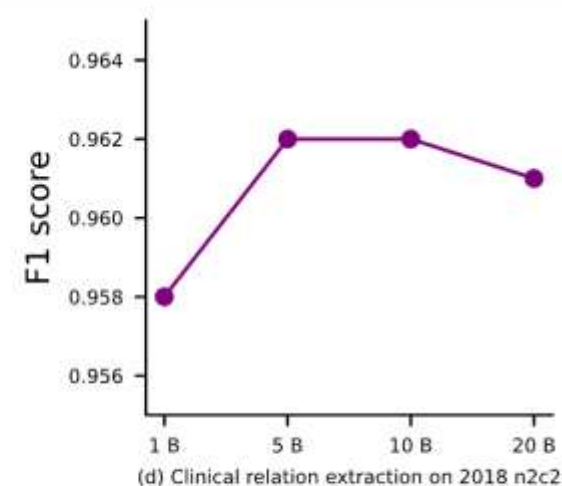
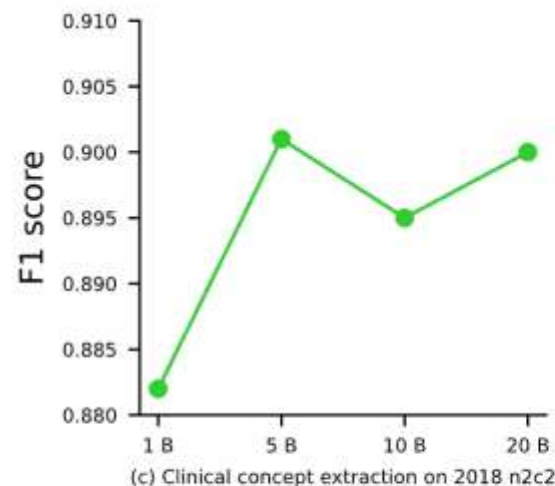
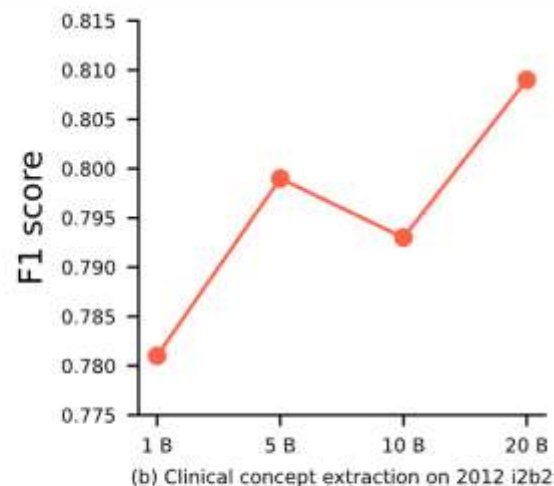
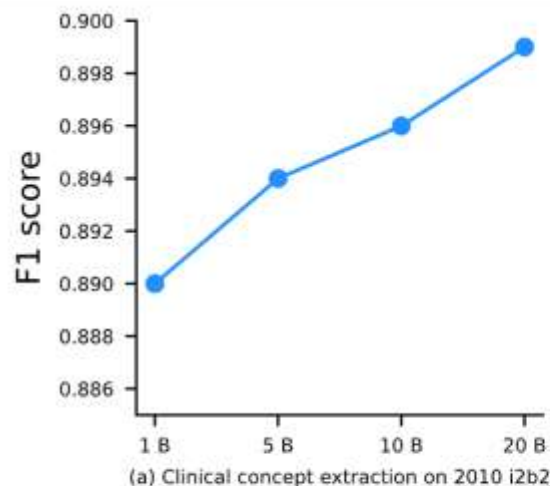


Synthetic NLP model - GatorTronS

	Clinical concept extraction									Medical relation extraction		
	2010 i2b2 ²⁰			2012 i2b2 ²¹			2018 n2c2 ²²			2018 n2c2 ²²		
Transformer	Precision	Recall	F1 score	Precision	Recall	F1 score	Precision	Recall	F1 score	Precision	Recall	F1 score
ClinicalBERT	NA	NA	0.878	NA	NA	0.789	0.859	0.883	0.871	0.968	0.941	0.954
GatorTron, 90B	0.875	0.904	0.889	0.764	0.822	0.792	0.876	0.904	0.890	0.972	0.948	0.960
GatorTronS, 1B	0.874	0.907	0.890	0.753	0.812	0.781	0.871	0.892	0.882	0.971	0.945	0.958
GatorTronS, 5B	0.879	0.909	0.894	0.777	0.823	0.799	0.899	0.903	0.901	0.974	0.949	0.962
GatorTronS, 10B	0.882	0.911	0.896	0.765	0.823	0.793	0.887	0.904	0.895	0.974	0.950	0.962
GatorTronS, 20B	0.889	0.911	0.899	0.784	0.836	0.809	0.892	0.907	0.900	0.975	0.947	0.961

Synthetic NLP model - GatorTronS

“synthetic” : “real notes” - unigrams (40.43 million : 4.82 million), bigrams (416.35 million : 62.51 million)



Turing test results

- No significant difference in linguistic readability ($p = 0.22$; 6.57 of GatorTronGPT compared with 6.93 of human)
- No significant difference in clinical relevance ($p = 0.91$; 7.0 of GatorTronGPT compared with 6.97 of human)
- Physicians cannot differentiate them ($p < 0.001$)

a. Percentage of notes correctly identified by human reviewers.			
	Ground truth		
	AI (n=30)	Human (n=30)	Total (n=60)
Physician 1	9 (30.0%)	17 (56.7%)	26 (43.3%)
Physician 2	13 (43.3%)	20 (66.7%)	33 (55%)
Overall	11 (36.7%)	18.5 (61.7%)	29.5 (49.2%)
p-value	< 0.001	= 0.104	< 0.001
b. Ratings of readability and clinical relevance by synthetic and human notes.			
	Ground truth		
	AI (n=30)	Human (n=30)	p-value
Readability	6.57 (1.22)	6.93 (1.09)	0.22
Clinical relevance	7.00 (1.23)	6.97 (1.07)	0.91
c.			
Example 1 (Passed Turing test): HISTORY OF PRESENT ILLNESS: Ms. [**NAME**] is a [**AGE**]-year-old female with prior history of left breast cancer who is referred to the [**LOCATION**] Clinic to discuss surgical management of her left lung nodule which was picked up on CT scanning for surveillance. The patient's chest CT obtained at an outside institution showed an incidental finding of a left lower lobe pleural nodule, which she brought with her to clinic today for further review. Ms. [**NAME**] reports to be doing well, with no change in her baseline level of shortness of breath, but denies any difficulty breathing, cough or hemoptysis. She feels essentially back to baseline and plays golf one day a week. Otherwise, Ms. [**NAME **] has no other pulmonary complaints and has never smoked tobacco or inhaled second hand smoke. She does report several years of intermittent reflux, which worsened after eating rough foods. She underwent an upper GI study for this, which showed a small hiatal hernia as well as a moderate- sized sliding hiatal hernia. She subsequently underwent an abdominal CT for evaluation for mesenteric ischemia and was felt to have a dilated pancreatic duct. Due to suspected head pathology on imaging (although the final report is pending), she underwent an MRI, which confirmed no pancreatic pathology. Given this, Ms. [**NAME **] has been trying to avoid fatty and fried foods, which has improved her reflux. Ms. [**NAME **] does report recent onset of sleep apnea, for which she is using CPAP every night for the last month. She tolerates this well without complaints of dry mouth, gasping for air, or undue sleepiness. Ms. [**NAME **] was seen in our outpatient [**LOCATION**] clinic on [**DATE**] at which time it was determined that in order to obtain a tissue diagnosis of this left lower lobe lesion a VATS approach would be utilized. She presents today for further pre-operative planning. Currently, Ms. [**DATE**] denies any pulmonary symptoms, fevers, chills, cough or weight loss. PAST MEDICAL HISTORY: 1. CARDIAC RISK FACTORS: Hypertension, Hyperlipidemia 2. CARDIAC HISTORY: None 3. OTHER PAST MEDICAL HISTORY: Left lower lobe pulmonary nodule			
Example 2 (Failed Turing test): Assessment: ICD-9-CM 1. Pain in joint, pelvic region and thigh, unspecified laterality 719.45 fentaNYL (DURAGESIC) 75 MCG/HR HYDROcodone-acetaminophen (VICODIN HP) 5-300 MG Tablet 2. Migraine NOS/not Intrcbl 346.90 SUMatriptan Succinate 3. Bone Pain Arthritis 333.90 Dexamethasone Sodium Phosphate 4. Bipolar disorder 296.80 traZODone (DESYREL) 100 MG Tablet prazosin (MINIPRESS) 2 MG capsule carvedilol (COREG) 3.125 MG tablet isosorbide mononitrate (IMDUR) 30 MG CR tablet Refer to Psychiatry clopidogrel (PLAVIX) 75 MG tablet SUMatriptan Succinate 5. ASTHMA UNSPECIFIED 493.90 albuterol (PROAIR HFA; VENTOLIN HFA) 108 (90 BASE) MCG/ACT inhaler 6. Major depressive disorder, single episode, unspecified 296.20 DULoxetine (CYMBALTA) 60 MG capsule Refer to Psychiatry amitriptyline (ELAVIL) 25 MG tablet traZODone (DESYREL) 100 MG Tablet 7. POST-SURGICAL VARICOSE VEINS OF LOWER EXTREMITIES 454.9 fentaNYL (DURAGESIC) 75 MCG/HR 8. Other and unspecified hyperlipidemia 272.4 simvastatin (ZOCOR) 40 MG tablet COMPREHENSIVE METABOLIC PANEL 9. PND (post-nasal drip) 784.91 loratadine (CLARITIN) 10 MG tablet 10. Bipolar I disorder, single manic episode, unspecified 296.00 clonazepam (Klonopin) 1 MG tablet Refer to Psychiatry 11. Allergic rhinitis 477.9 loratadine (CLARITIN) 10 MG tablet 12. Grief reaction 309.0 traZODone (DESYREL) 100 MG Tablet 13. Encounter for long-term (current) use of other medications V58.69 methocarbamol (ROBAXIN) 750 MG tablet COMPREHENSIVE METABOLIC PANEL 14. GERD (gastroesophageal reflux disease) 530.81 lansoprazole (PRE			

Written by UF Health Physician

Paragraph 1

HISTORY OF PRESENT ILLNESS: Ms. [**NAME**] is a [**AGE**]-year-old female with prior history of left breast cancer who is referred to our department for evaluation for radiation as part of breast-conserving therapy of a newly diagnosed cancer in her right breast. Information pertinent to this dictation as follows. In [**DATE**], Ms. [**NAME**] was diagnosed and treated for cancer of her left breast at [**LOCATION**]. We do not have full outside records related to her diagnosis and treatment for left breast cancer. However, she reportedly underwent lumpectomy followed by a sentinel node biopsy in [**LOCATION**]. She claims the sentinel node biopsy did not show evidence of cancer and she then underwent adjuvant radiation and chemotherapy at [**LOCATION**]. Her radiation treatment consisted of approximately six weeks of daily radiation therapy targeting her left breast. She is unsure of the exact details of her chemotherapy, its agents, or the number of courses. Following her completion of chemotherapy, she was then treated with Arimidex which she took for a total of five years. She followed locally in the [**LOCATION**] area at the [**LOCATION**] Clinic for approximately eight years after treatment, without any evidence of recurrent breast cancer. In late [**DATE**], she decided to relocate along with her husband to [**LOCATION**]. She is currently in the process of transferring her care to this region but her outside physicians were still in the [**LOCATION**] area so she decided to return for her follow-up mammograms to [**LOCATION**]. On [**DATE**], she went back to [**LOCATION**] in [**LOCATION**], for routine bilateral diagnostic mammogram. Prior to this, her earlier mammograms in [**DATE**] and [**DATE**] had both times been read as benign BI-RADS 2. On this date, her mammogram demonstrated benign-appearing left breast with no evidence of malignancy. However, the mammogram of her right breast demonstrated on spot compression images, the persistence of a focal density in the upper outer region. She also underwent directed ultrasound evaluation, which demonstrated a lobulated hypoechoic nodule in the 9 o'clock position measuring 1.63 x 0.83 x 1.12 cm, for which biopsy was recommended. No additional masses were seen in the axilla.

Written by GatorTronGPT

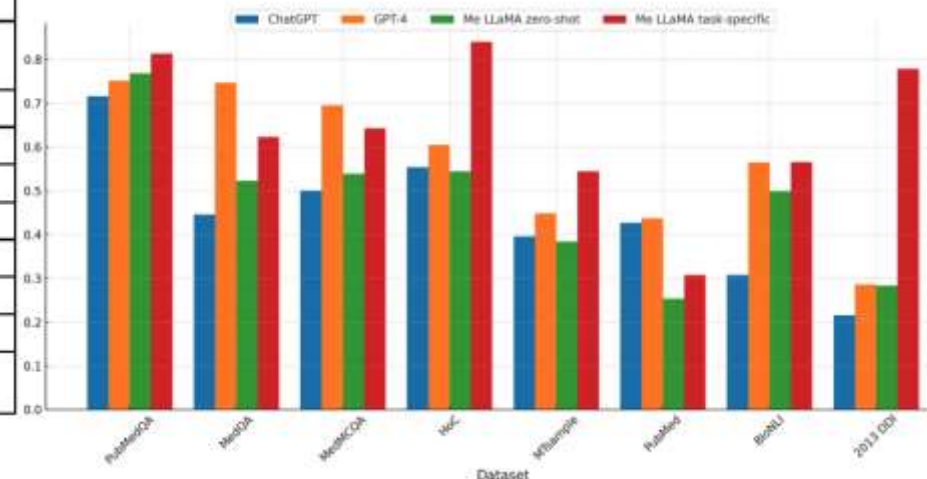
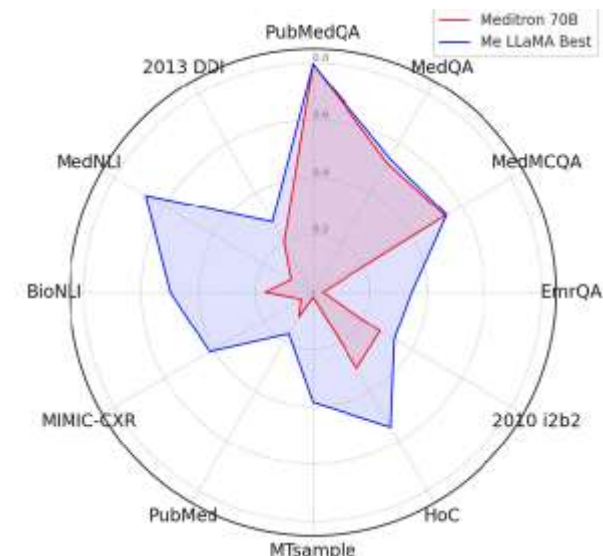
Paragraph 2

HISTORY OF PRESENT ILLNESS: Ms. [**NAME**] is a [**AGE**]-year-old female with prior history of left breast cancer who is referred to [**LOCATION**] to discuss surgical management of her left lung nodule which was picked up on CT scanning for surveillance. The patient's chest CT obtained at an outside institution showed an incidental finding of a left lower lobe pleural nodule, which she brought with her to clinic today for further review. Ms. [**Name**] reports to be doing well, with no change in her baseline level of shortness of breath, but denies any difficulty breathing, cough or hemoptysis. She feels essentially back to baseline and plays golf one day a week. Otherwise, Ms. [**NAME**] has no other pulmonary complaints and has never smoked tobacco or inhaled second hand smoke. She does report several years of intermittent reflux, which worsened after eating rough foods. She underwent an upper GI study for this, which showed a small hiatal hernia as well as a moderate- sized sliding hiatal hernia. She subsequently underwent an abdominal CT for evaluation for mesenteric ischemia and was felt to have a dilated pancreatic duct. Due to suspected head pathology on imaging (although the final report is pending), she underwent an MRI, which confirmed no pancreatic pathology. Given this, Ms. [**NAME**] has been trying to avoid fatty and fried foods, which has improved her reflux. Ms. [**NAME**] does report recent onset of sleep apnea, for which she is using CPAP every night for the last month. She tolerates this well without complaints of dry mouth, gasping for air, or undue sleepiness. Ms. [**NAME**] was seen in our [**LOCATION**] clinic on [**DATE**] at which time it was determined that in order to obtain a tissue diagnosis of this left lower lobe lesion a VATS approach would be utilized. She presents today for further pre-operative planning. Currently, Ms. [**DATE**] denies any pulmonary symptoms, fevers, chills, cough or weight loss. PAST MEDICAL HISTORY: 1. CARDIAC RISK FACTORS: Hypertension, Hyperlipidemia 2. CARDIAC HISTORY: None 3. OTHER PAST MEDICAL HISTORY: Left lower lobe pulmonary nodule

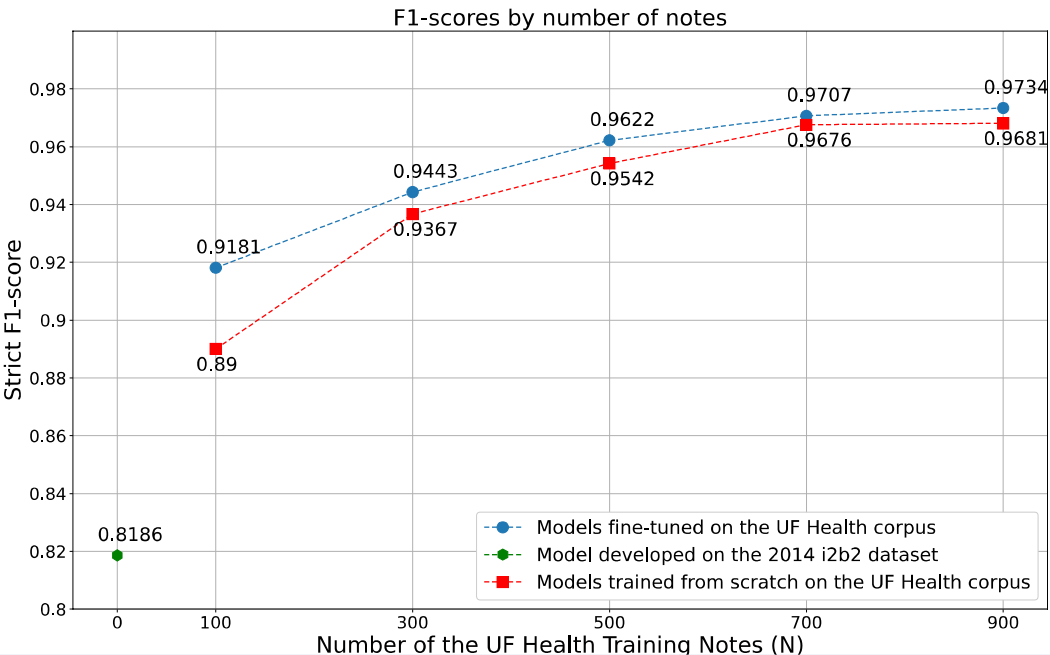
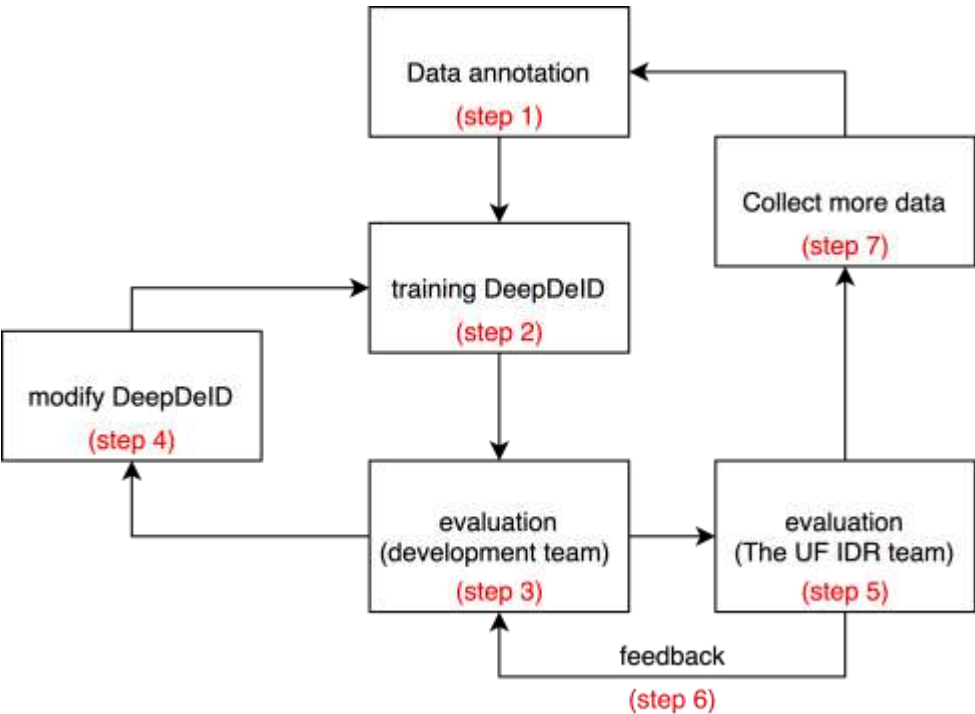
Me-LLaMA: Foundation Large Language Models for Medical Applications

Qianqian Xie , Qingyu Chen , Aokun Chen , Cheng Peng , Yan Hu , Fongci Lin , Xueqing Peng , Jimin Huang , Jeffrey Zhang , Vipina Keloth , Xinyu Zhou , Huan He , Lucila Ohno-Machado , Yonghui Wu , Hua Xu , Jiang Bian 

Task	Dataset	Metric	LLaMA2-13B-chat	PMC-LLaMA-chat	Medalpa-ca-13B	AlpaCar-e-13B	Me-LLaMA 13B-chat	LLaMA2-70B-chat	Meditr on 70B	Me-LLaMA 70B-chat
Question answering	PubMedQA	Accuracy	0.546	0.504	0.238	0.538	0.700	0.668	0.718	0.768
		Macro-F1	0.457	0.305	0.192	0.373	0.504	0.477	0.516	0.557
	MedQA	Accuracy	0.097	0.207	0.143	0.304	0.427	0.376	0.428	0.523
		Macro-F1	0.148	0.158	0.102	0.281	0.422	0.367	0.419	0.521
	MedMCQA	Accuracy	0.321	0.212	0.205	0.385	0.449	0.339	0.368	0.539
		Macro-F1	0.243	0.216	0.164	0.358	0.440	0.273	0.382	0.538
	EmrQA	Accuracy	0.001	0.053	0.000	0.001	0.048	0.050	0.000	0.119
		F1	0.098	0.304	0.040	0.198	0.307	0.251	0.000	0.346
Named entity recognition	2010 i2b2	Macro-F1	0.143	0.091	0.000	0.173	0.166	0.321	0.121	0.329
Relation extraction	2013 DDI	Macro-F1	0.090	0.147	0.058	0.110	0.214	0.087	0.176	0.283
Classification	HoC	Macro-F1	0.228	0.184	0.246	0.267	0.335	0.309	0.258	0.544
	MTsample	Macro-F1	0.133	0.083	0.003	0.273	0.229	0.254	0.142	0.384
Summarization	PubMed	Rouge-L	0.161	0.028	0.014	0.167	0.116	0.192	0.169	0.169
		BERTS*	0.671	0.128	0.117	0.671	0.445	0.684	0.658	0.678
	MIMIC-CXR	Rouge-L	0.144	0.139	0.010	0.134	0.400	0.131	0.060	0.418
		BERTS*	0.704	0.694	0.502	0.702	0.797	0.696	0.582	0.787
Natural language inference	BioNLI	Macro-F1	0.173	0.159	0.164	0.170	0.195	0.297	0.194	0.436
	MedNLI	Macro-F1	0.412	0.175	0.175	0.275	0.472	0.515	0.218	0.675



UF CTSI/IDR De-identification

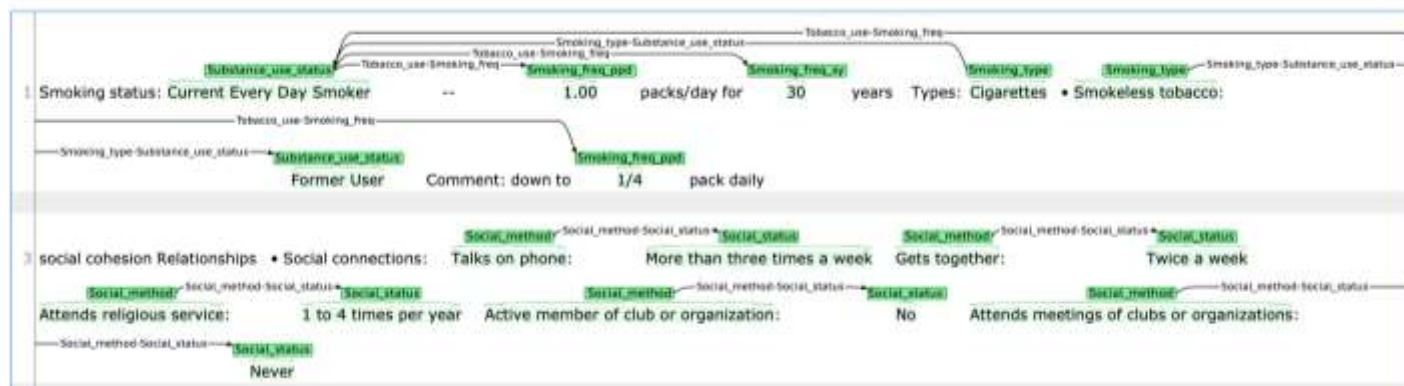


System	Performance		
	Precision	Recall	F1-score
Commercial product	0.8878	0.9160	0.9017
DeepDeID (v1.7)	0.9657	0.9555	0.9606
DeepDeID (v1.8)	0.9627	0.9976	0.9798

Yang X, Lyu T, Li Q, Lee C-Y, Bian J, Hogan WR, **Wu Y†**. A study of deep learning methods for de-identification of clinical notes in cross-institute settings. *BMC Med Inform Decis Mak*. 2020 Dec 5;19(5):232. <https://www.ncbi.nlm.nih.gov/pubmed/31801524>.

Yang X, Bian J, **Wu Y†**. Customize Deep Learning-based De-Identification Systems Using Local Clinical Notes - A Study of Sample Size *KDD*. 2020. In press.

UF CTSI/NLP Core SDoH service



Financial Resource Strain Difficulty paying for basics: Not hard at all

Marital status Poverty

"She is a widow very limited income"

Social Determinants of Health



Social Determinants of Health
Copyright-free
Healthy People 2030

Support: Unemployment: Other: Income Usually: X Meets Expenses Exceeds Expenses Is less than

*Financial_constrain *Sdoh_status*

expenses Medical Insurance: Medicaid: (specify) X Private Plan:(specify) Cigna; via mother's employment TBD Insurance includes

• TUBAL LIGATION Social History: She reports that she has never smoked. She has never used smokeless tobacco. She

reports that she does not drink alcohol. Marital Status:{Desc; marital status:2100062} Children:{gen number 0-10:310397}

SODA



Original Research

Identifying social determinants of health from clinical narratives: A study of performance, documentation ratio, and potential bias

Zehao Yu^a, Cheng Peng^{a,b}, Xi Yang^{a,b}, Chong Dang^a, Prakash Adekkanattu^c, Braja Gopal Patra^d, Yifan Peng^d, Jyotishman Pathak^d, Debbie L. Wilson^e, Ching-Yuan Chang^e, Wei-Hsuan Lo-Ciganic^e, Thomas J. George^f, William R. Hogan^a, Yi Guo^{a,b}, Jiang Bian^{a,b}, Yonghui Wu^{a,b,*}



https://github.com/uf-hobi-informatics-lab/SODA_Docker

The documentation ratios of the Black group for “**Abuse**”, “**Financial constraint**”, “**Living condition**”, “**Physical activity**”, “**Social cohesion**”, and “**Transportation**” are remarkably higher than other groups (over 10%)”, reflecting findings from literature:

- Black children were more than twice as likely to be referred for abused victims;
- Black older adults are at heightened risk of overall mistreatment and financial abuse;
- Discrimination towards Black individuals in rental and housing markets remains pervasive;
- Higher work-related physical activity among Black compared with White;
- African American has more transportation burden than other groups

SDoH	White N = 15,543	Black N = 3,289	Other N = 994	Male N = 8,216	Female N = 12,972
Abuse (physical or mental)	0.42	0.55	0.34	0.40	0.46
Alcohol use	0.95	0.98	0.91	0.94	0.95
Drug use	0.92	0.96	0.88	0.91	0.92
Education	0.88	0.95	0.83	0.87	0.89
Ethnicity	0.65	0.70	0.60	0.56	0.69
Financial constraint	0.30	0.42	0.28	0.27	0.35
Gender	0.99	0.99	0.97	0.98	0.99
Language	0.64	0.70	0.58	0.55	0.69
Living condition	0.50	0.64	0.46	0.52	0.51
Living supply	0.84	0.91	0.79	0.82	0.86
Marital status	0.91	0.96	0.89	0.89	0.91
Occupation/ Employment	0.88	0.92	0.86	0.86	0.89
Physical activity	0.35	0.47	0.31	0.30	0.41
Race	0.82	0.88	0.56	0.76	0.83
SDoH ICD	0.09	0.19	0.06	0.12	0.10
Sexual activity	0.85	0.92	0.78	0.83	0.85
Social cohesion	0.30	0.43	0.26	0.27	0.35
Tobacco use	0.75	0.84	0.69	0.77	0.75
Transportation	0.31	0.43	0.28	0.28	0.36

	(±0.003)	(±0.006)	(±0.004)	(±0.003)	(±0.007)	(±0.004)
Longformer	0.9625	0.9498	0.9561	0.9635	0.9507	0.9571
	(±0.005)	(±0.007)	(±0.004)	(±0.005)	(±0.007)	(±0.004)
GatorTron	0.9183	0.8754	0.8963	0.9356	0.8919	0.9133

SDoH affect Breast Cancer Survival?

- Protective SDoH
 - Active physical activity
- Risk SDoH
 - Divorced
 - Single

PCORI ME-2018C3-14754 - NLP to Connect Social Determinants and Clinical Factors for Outcomes Research. **(Wu, Bian)**

Variables	HR	CI lower	CI upper	P value	
Ethnitic:Not Hispanic or Latino	1.999	0.703	5.683	0.194	
Ethnitic:Other	7.867	1.929	32.080	0.004	**
Race:Asian	0.705	0.260	1.914	0.493	
Race:Black	1.695	1.284	2.238	<0.001	***
Race:Other	0.681	0.383	1.212	0.191	
Age:30-40	0.700	0.326	1.501	0.359	
Age:40-50	0.591	0.290	1.202	0.146	
Age:50-60	1.088	0.547	2.162	0.811	
Age:60-70	1.455	0.725	2.921	0.292	
Age:70-80	2.119	1.026	4.377	0.042	*
Age:80+	3.389	1.488	7.720	0.004	**
Cancer stage:Regional	1.274	0.508	3.192	0.605	
Cancer stage:Distant	3.100	1.319	7.289	0.009	**
Cancer stage:Other	0.993	0.479	2.060	0.985	
Education:College or above	0.645	0.398	1.047	0.076	.
Education:Miss	0.725	0.503	1.046	0.086	.
Physical active:Physically Active	0.514	0.279	0.947	0.033	*
Physical active:Miss	1.355	0.881	2.084	0.167	
Marital status:Single	1.840	1.373	2.464	<0.001	***
Marital status:Widow	0.949	0.672	1.341	0.768	
Marital status:Divorced	1.794	1.304	2.468	<0.001	***
Marital status:Miss	0.573	0.319	1.031	0.063	.
Social cohesion:Yes	0.760	0.282	2.049	0.587	
Social cohesion:Miss	5.264	2.779	9.970	<0.001	***
Employment status:Retired/disabled	0.862	0.620	1.198	0.376	
Employment status:Unemployed	1.147	0.789	1.668	0.473	
Employment status:Miss	0.908	0.683	1.207	0.506	
Drug use:Yes	0.881	0.520	1.494	0.638	
Drug use:Miss	0.639	0.397	1.031	0.066	.
Abuse:Abused	0.468	0.281	0.780	0.004	**
Abuse:Miss	0.667	0.465	0.957	0.028	*

The diagram illustrates the semantic network of a thyroid ultrasound report. It maps specific terms from the report to their corresponding semantic classes, represented by colored dots and labels. The report is divided into several sections, each describing a different part of the thyroid gland or associated structures.

Top Section: Large Nodule in the Right Lobe

- large**: linked to *size_qualitative* (red dot).
- well-defined**: linked to *margins* (purple dot).
- isoechoic**: linked to *echogenicity* (red dot).
- and heterogeneous**: linked to *composition* (green dot).
- nodule**: linked to *thyroid_nodule* (blue dot).
- occupying much of**: linked to *size* (green dot).
- the RIGHT lobe**: linked to *location* (purple dot).
- This measures 2.6 x 2.2 x 2.3 cm**: linked to *size_numeric* (green dot).
- demonstrates mild increase in vascularity**: linked to *size_qualitative* (red dot).

Middle Section: Small Hypoechoic Nodule in the Left Lobe

- small**: linked to *size_qualitative* (red dot).
- hypoechoic**: linked to *echogenicity* (red dot).
- nodule**: linked to *thyroid_nodule* (blue dot).
- measuring 7.8 x 5.6 x 6.8 mm**: linked to *size_numeric* (green dot).
- There is a cystic and solid-appearing**: linked to *composition* (green dot).
- nodule**: linked to *thyroid_nodule* (blue dot).
- measuring**: linked to *size* (green dot).

Bottom Section: Nodules in the Medial Left Lobe, Isthmus, and Lymph Nodes

- 1 x 1 x 1.4 cm. LEFT THYROID LOBE:** linked to *size_numeric* (green dot).
- 4.7 x 1.9 x 1.8 cm. Thyroid echogenicity: Heterogeneous echogenicity throughout. Thyroid vascularity: Moderately increased vascularity. Presence of thyroid**: linked to *size_numeric* (green dot).
- nodules: There is a hypoechoic**: linked to *echogenicity* (red dot).
- well-defined**: linked to *margins* (purple dot).
- nodule**: linked to *thyroid_nodule* (blue dot).
- in the medial LEFT lobe measuring 1.4 x 0.9 x 1.1 cm**: linked to *size_numeric* (green dot).
- ISTHMUS:** linked to *size* (green dot).
- 0.28 cm**: linked to *size_numeric* (green dot).
- LYMPH NODES: Lymph node**: linked to *size* (green dot).

Site location: Upper lobe Tumor Size: Greatest dimension (cm): 4.5 Additional dimensions (cm): 3.5 x 2.5 Tumor Focality: Unifocal Histologic Type: Squamous cell carcinoma Histologic Grade: G2:

Moderately differentiated Visceral Pleura Invasion: Not identified Tumor Extension: Intraparenchymal only. Margins: Bronchial Margin: Uninvolved by invasive carcinoma Vascular Margin: Uninvolved by invasive carcinoma

ttf1 positive lung adenocarcinoma with large necrotic right adrenal gland **metex14** skipping **c.2942-19_2949del** 2. may 1
 ● biomarker* ● biomarker* ● *Mutation

8/2/18 egfr neg, pdl-1 1%, ros1 neg, her2 neg, alk neg, ret neg, met neg,

● biomarker* ● biomarker* ● biomarker* ● biomarker* ● biomarker* ● biomarker*

Predictive model

- Readmission
- Mortality
- Comorbidity
- Length of stay
- Insurance denial

Article

Health system-scale language models are all-purpose prediction engines

<https://doi.org/10.1038/s41586-023-06160-y>

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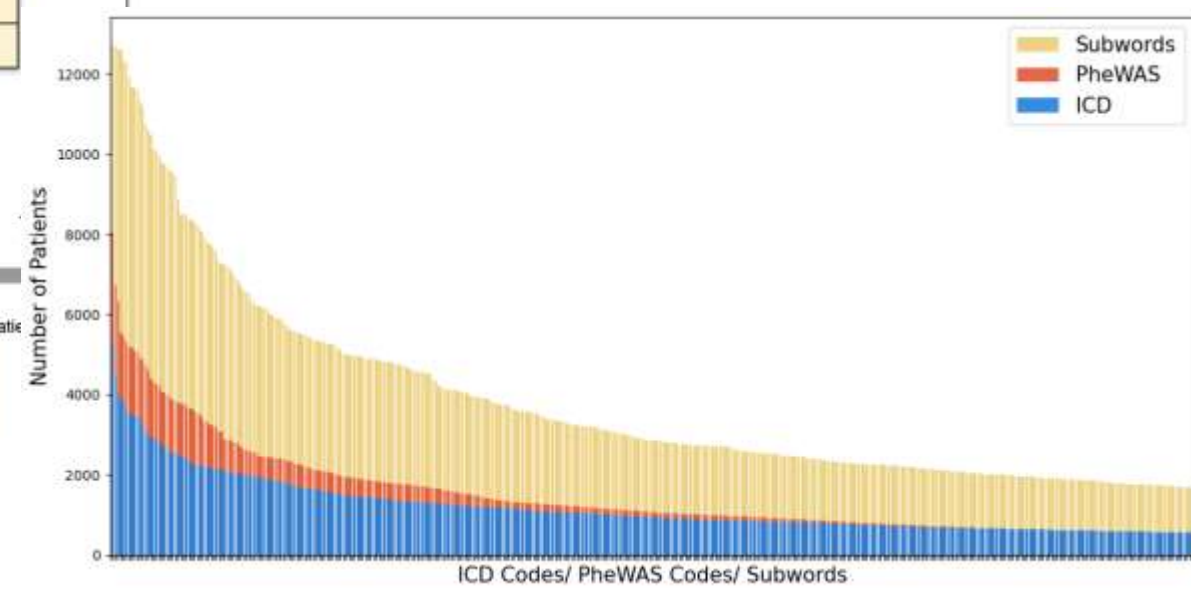
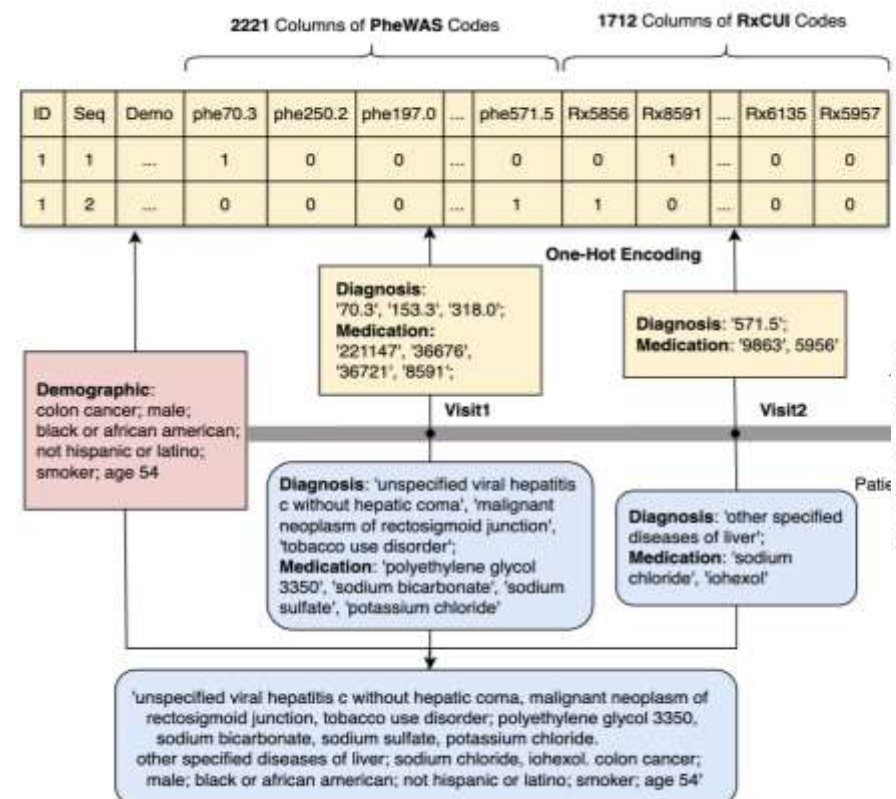
Open access

 Check for updates

Lavender Yao Jiang^{1,2}, Xujin Chris Liu^{1,3}, Nima Pour Nejatian⁴, Mustafa Nasir-Moin¹, Duo Wang⁵, Anas Abidin⁴, Kevin Eaton⁶, Howard Antony Riina¹, Ilya Laufer¹, Paawan Punjabi⁶, Madeline Miceli⁶, Nora C. Kim¹, Cordelia Orillac¹, Zane Schnurman¹, Christopher Livia¹, Hannah Weiss¹, David Kurland¹, Sean Neifert¹, Yosef Dastagirzada¹, Douglas Kondziolka¹, Alexander T. M. Cheung¹, Grace Yang^{1,2}, Ming Cao^{1,2}, Mona Flores⁴, Anthony B. Costa⁴, Yindalon Aphinyanaphongs^{5,7}, Kyunghyun Cho^{2,8,9,10} & Eric Karl Oermann^{1,2,11}✉

Physicians make critical time-constrained decisions every day. Clinical predictive models can help physicians and administrators make decisions by forecasting clinical and operational events. Existing structured data-based clinical predictive models have limited use in everyday practice owing to complexity in data processing, as well as model development and deployment^{1–3}. Here we show that unstructured clinical notes from the electronic health record can enable the training of clinical language models, which can be used as all-purpose clinical predictive engines with low-resistance development and deployment. Our approach leverages recent advances in natural language processing^{4,5} to train a large language model for medical language (NYUTron) and subsequently fine-tune it across a wide range of clinical and operational predictive tasks. We evaluated our approach within our health system for five such tasks: 30-day all-cause readmission prediction, in-hospital mortality prediction, comorbidity index prediction, length of stay prediction, and insurance denial prediction. We show that NYUTron has an area under the curve (AUC) of 78.7–94.9%, with an improvement of 5.36–14.7% in the AUC compared with traditional models. We additionally demonstrate the benefits of pretraining with clinical text, the potential for increasing generalizability to different sites through fine-tuning and the full deployment of our system in a prospective, single-arm trial. These results show the potential for using clinical language models in medicine to read alongside physicians and provide guidance at the point of care.

Assess Disease Risk



Model	Feature encoding	F1	Precision	Recall	AUC	Specificity	Accuracy
XGB	One-hot	0.290	0.592	0.192	0.825	0.982	0.889
SVMs	One-hot	0.302	0.376	0.252	0.756	0.944	0.863
T-LSTM	Sequence	0.622	0.650	0.596	0.907	0.957	0.914
BERT	Subword	0.643	0.631	0.656	0.890	0.949	0.914
GatorTron-345M	Subword	0.692	0.716	0.669	0.930	0.965	0.929
GatorTron-3.9B	Subword	0.699	0.684	0.715	0.896	0.956	0.927



Ziyi Chen, MS

Reduce Physician's Documentation Burden

Automatic Summarization of Doctor-Patient Encounter Dialogues Using Large Language Model through Prompt Tuning.

Mengxian Lyu, Cheng Peng, Xiaohan Li, Patrick Balian, Jiang Bian, Yonghui Wu



Mengxian Lyu, MS

‘We’re getting much more aggressive’: Microsoft’s Nuance adds GPT-4 AI to its medical note-taking tool

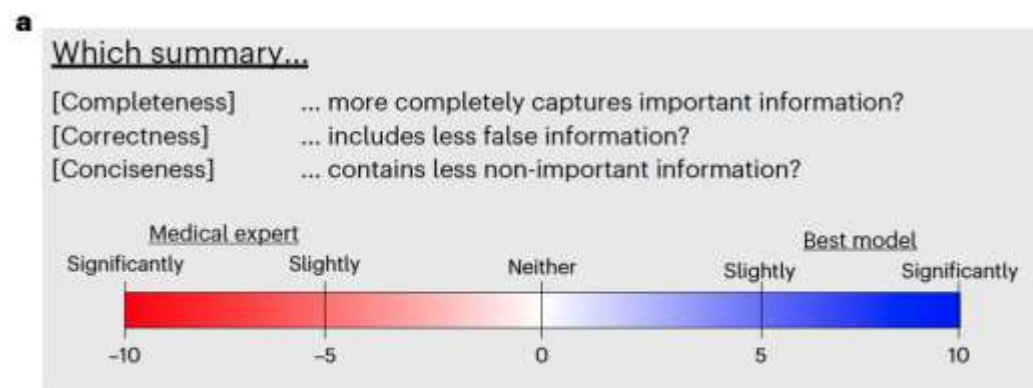


By [Brittany Trang](#) and [Casey Ross](#) March 20, 2023

[Reprints](#)

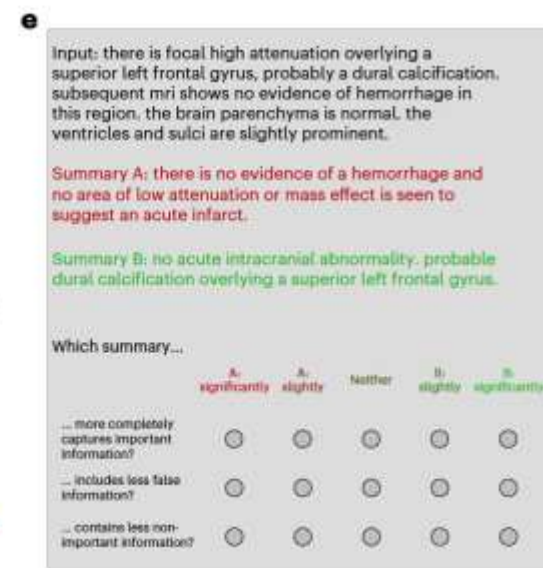
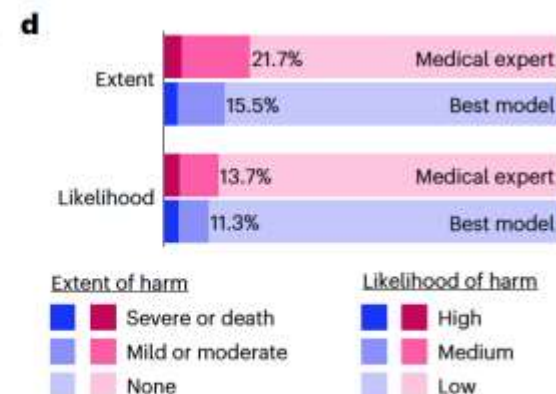
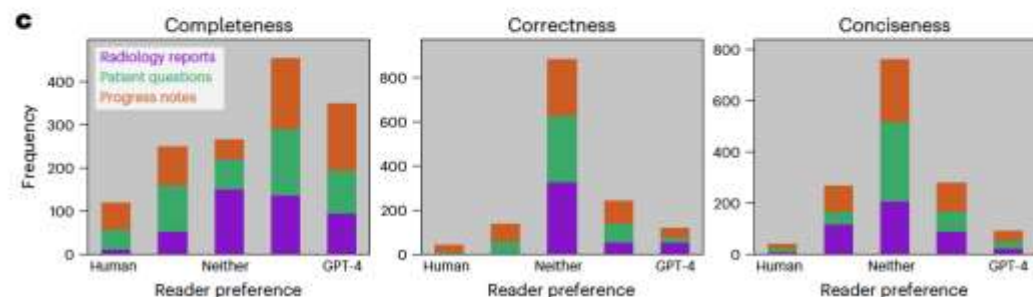
Doctor-Patient Dialogue	Clinical Note Section
Doctor: What brings you back into the clinic today, miss? Patient: I came in for a refill of my blood pressure medicine. Doctor: It looks like Doctor [**{NAME}**] followed up with you last time regarding your hypertension, osteoarthritis, osteoporosis, hypothyroidism, allergic rhinitis and kidney stones. Have you noticed any changes or do you have any concerns regarding these issues? Patient: No. Doctor: Have you had any fever or chills, cough, congestion, nausea, vomiting, chest pain, chest pressure? Patient: No. Doctor: Great. Also, for our records, how old are you and what race do you identify yourself as? Patient: I am seventy six years old and identify as a white female.	The patient is a 76-year-old white female who presents to the clinic today originally for hypertension and a med check. She has a history of hypertension, osteoarthritis, osteoporosis, hypothyroidism, allergic rhinitis and kidney stones. Since her last visit she has been followed by [**{NAME}**]. Those issues are stable. She has had no fever or chills, cough, congestion, nausea, vomiting, chest pain, chest pressure.

Adapted large language models can outperform medical experts in clinical text summarization



b

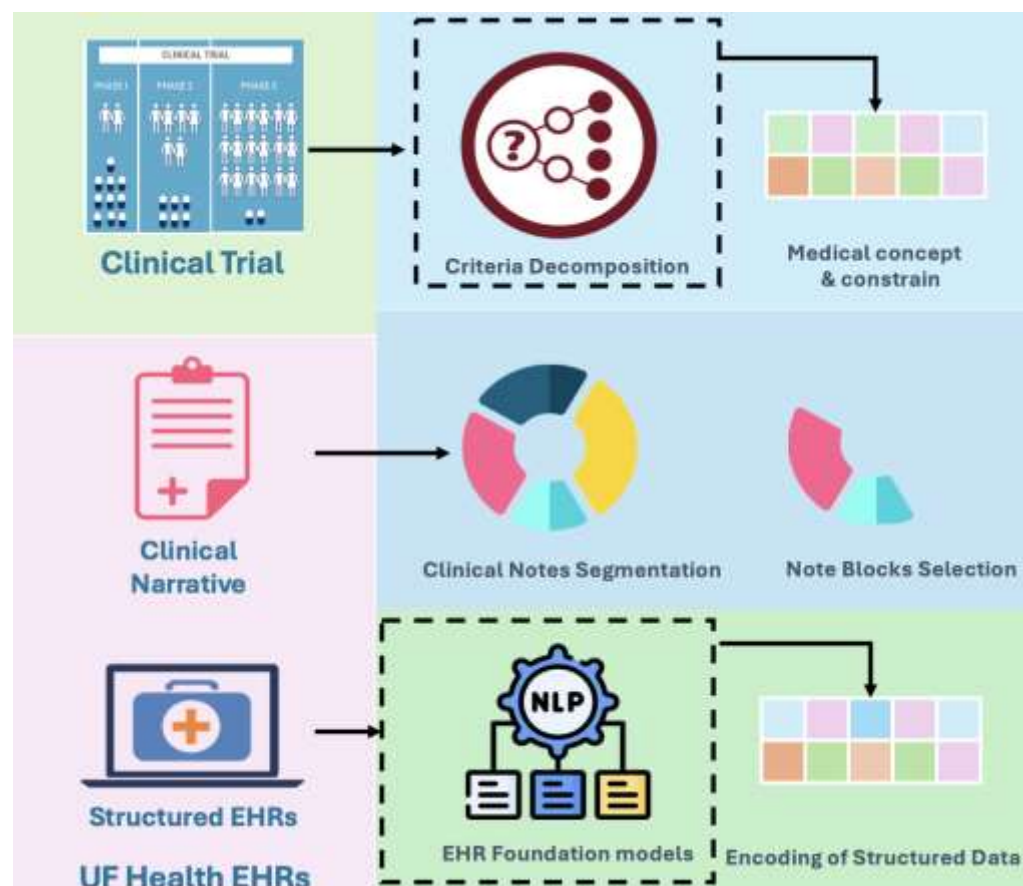
Task	Completeness	Correctness	Conciseness
Radiology reports	2.8 ± 5.1 *	1.7 ± 3.7 *	0 ± 4.3
Patient questions	1.6 ± 6.5 *	0.6 ± 3.7 *	0.6 ± 3.9 *
Progress notes	2.6 ± 6.9 *	0.4 ± 4.8	0.6 ± 4.5 *
Overall	2.3 ± 5.8 *	0.8 ± 3.7 *	0.4 ± 4.0 *



Clinical trial recruitment



Aokun Chen, PhD
Assistant Scientist



Criteria:

Patients with uncontrolled hypertension, history of significant liver disease, clinically significant pulmonary disease, diabetes, or cancer will be excluded

...

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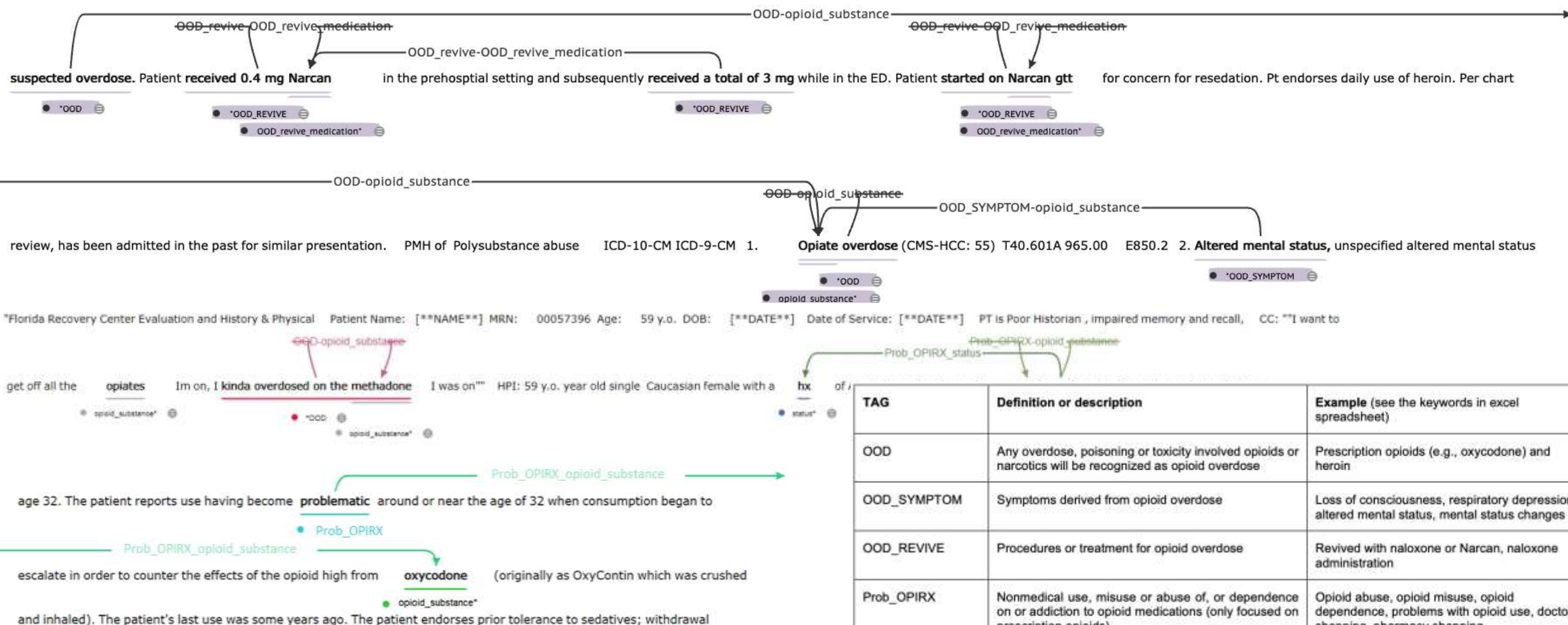


Prompt:

Context: "context": The clinical notes-based answer to the question whether this patient have <|CORE_CONCEPT|> fulfill the conditions for <|CONSTRAIN|> includes <NUM> Yes, <NUM> No, <NUM> unknown
question: Does this patient have <|CORE_CONCEPT|> fulfill the conditions for <|CONSTRAIN|>?

Evaluate using a UF Health Alzheimer's trial, 100 patients:

Precision:0.88 Recall:0.85 F1 score: 0.86



Medication Exposures and Changes



Contextualized medication information extraction using Transformer-based deep learning architectures

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^a Department of Health Outcomes and Biomedical Informatics, College of Medicine, University of Florida, Gainesville, FL, USA

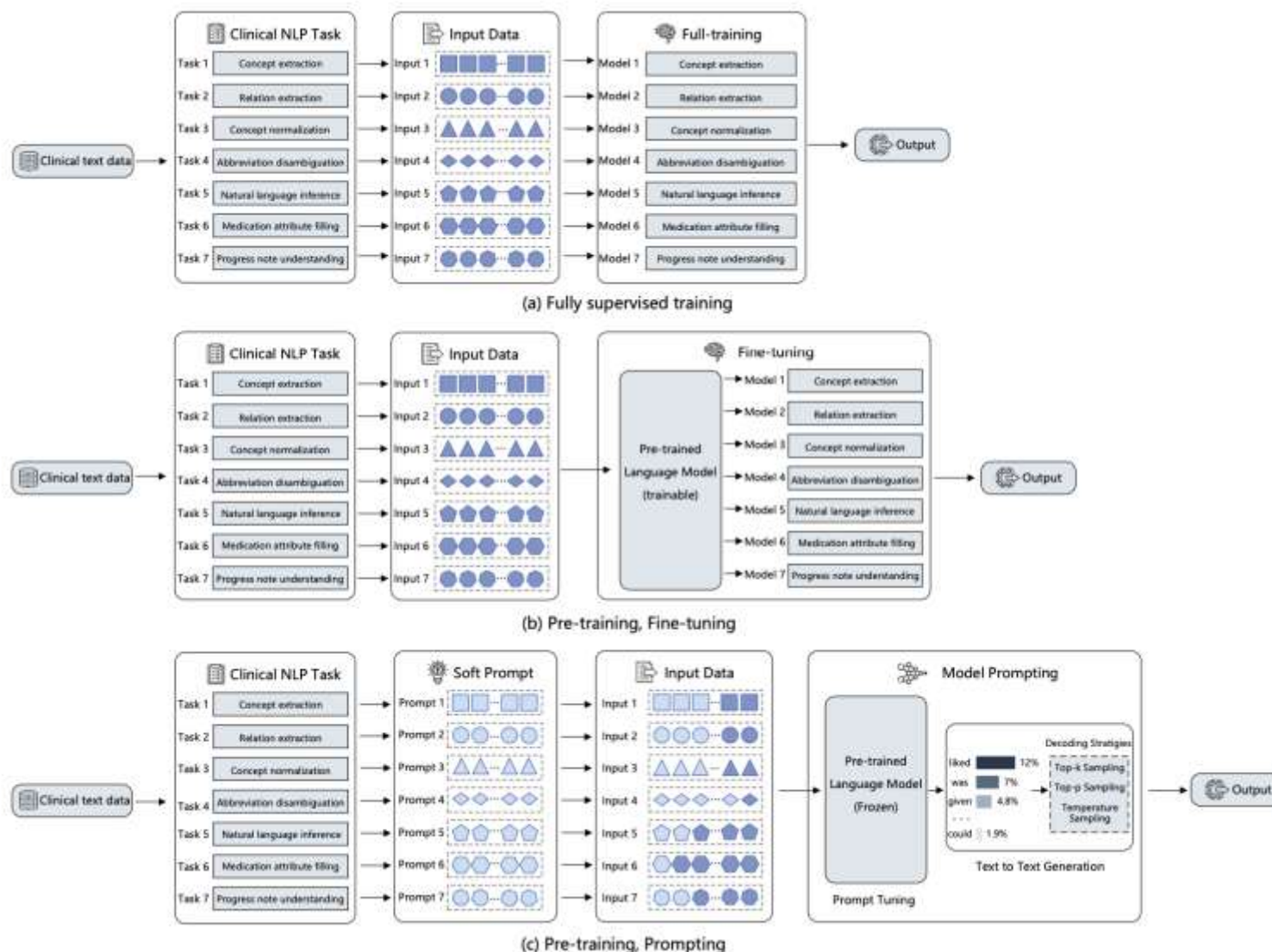
^b Cancer Informatics Shared Resource, University of Florida Health Cancer Center, Gainesville, FL, USA

Task	Categories	Count in train (%)	Count in test (%)	Task	Categories	Count in train (%)	Count in test (%)
Event	No Disposition	5260 (72.8%)	1326 (74.4%)	Temporality	Past	744 (52.7%)	173 (51.6%)
	Disposition	1412 (19.5%)	335 (18.9%)		Present	494 (35.0%)	132 (39.4%)
	Undetermined	557 (7.7%)	122 (6.8%)		Future	145 (10.3%)	29 (8.7%)
Action	Start	568 (40.2%)	131 (39.1%)	Certainty	Unknown	29 (2.1%)	1 (0.3%)
	Stop	340 (24.1%)	67 (20.0%)		Certain	1176 (83.3%)	281 (83.9%)
	Increase	129 (9.1%)	22 (6.6%)		Hypothetical	134 (9.5%)	33 (9.9%)
	Decrease	54 (3.8%)	13 (3.9%)		Conditional	100 (7.1%)	15 (4.5%)
	Unique Dose	285 (20.2%)	88 (26.3%)				
	Other Change	1 (0.1%)	0 (0.0%)				
	Unknown	35 (2.5%)	14 (4.2%)				
Negation	Negated	32 (2.3%)	6 (1.8%)	Actor			
	Not Negated	1380 (97.7%)	329 (98.2%)				
Medication extrac				Medication extrac	Model	Accuracy	
						Overall	Action
							Negation
							Temporal
							Certainty
							Actor
Model	Micro			Model	Accuracy		
					Precision		
					Recall		
					F1		
GatorTron	0.9772			GatorTron	0.9887		
					0.9743		
					0.8772		
					0.9801		
					0.8673		
					0.9653		
GatorTronS	0.9840			GatorTronS	0.9828		
					0.9791		
					0.8759		
					0.9816		
					0.8571		
					0.9663		
RoBERTa	0.8746			RoBERTa	0.9828		
					0.9791		
					0.8759		
					0.9816		
					0.8571		
					0.9663		
RoBERTa_MIMIC	0.9832			RoBERTa_MIMIC	0.9828		
					0.9791		
					0.8759		
					0.9816		
					0.8571		
					0.9663		
ALBERT	0.8472			ALBERT	0.9828		
					0.9791		
					0.8759		
					0.9816		
					0.8571		
					0.9663		
ALBERT_MIMIC	0.9673			ALBERT_MIMIC	0.9828		
					0.9791		
					0.8759		
					0.9816		
					0.8571		
					0.9663		

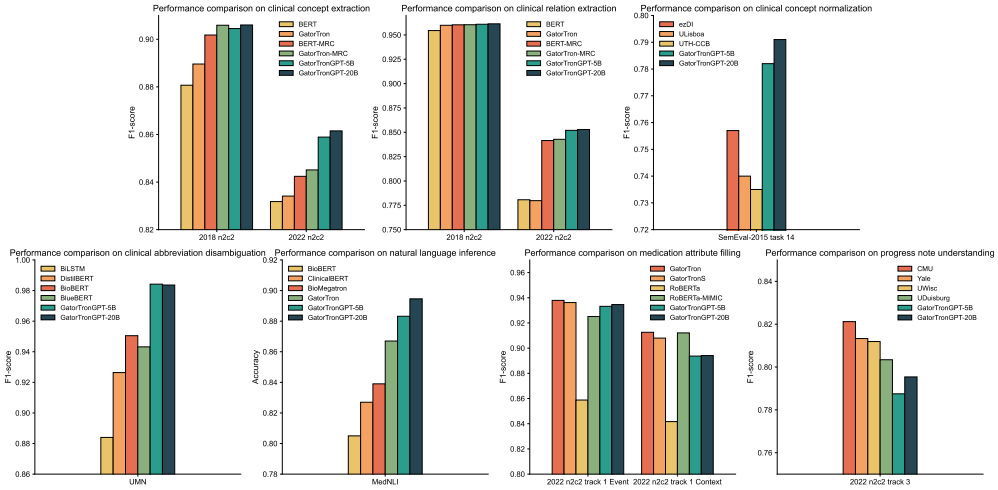
LLM Prevails

- Convenient for general users – use without coding
- A general form of AI – one model solving many tasks
- Free text question answering (QA) systems
- Generative
- Transfer learning
- Few-shot learning
- Zero-shot learning

Prevail-One model solves all NLP tasks



1. Nonlogical hallucination: the LLM generated sequences without logic.	Input data Disp: * 60 80 mg syringe * Refills: * 1 * 8. [s2] Acetaminophen [e2] 500 mg Tablet Sig: 1 - 2 Tablets PO [s1] Q6H (every 6 hours) as needed [e1] for pain.	Ground truth The relation between “Q6H (every 6 hours) as needed” and “Acetaminophen” is “ Frequency-Drug ”.	Hallucination - Non-responder - Non-responder - Non-responder
	8. docusate sodium 100 mg [s1] Capsule [e1] Sig: One (1) Capsule PO BID (2 times a day). 7. [s2] senna [e2] 8.6 mg Tablet Sig: One (1) Tablet PO BID (2 times a day) as needed for constipation.	The relation between “Capsule” and “senna” is “ No-relation ”.	1: 2: 3: 4: 5: 6: 7: 8: 9: 10: 11: 12: 13: 14;
2. Irrelevant hallucination: the LLM generated readable but irrelevant answers.	Disp: * 30 Tablet (s) * Refills: * 0 * 24. [s2] clopidogrel [e2] 75 mg Tablet Sig: [s1] One (1) [e1] Tablet PO DAILY (Daily)	The relation between “One (1)” and “clopidogrel” is “ Dosage-Drug ”.	1. The drug clopidogrel is a non-steroidal anti-platelet drug that is used to prevent platelet aggregation and reduce the risk of thrombosis in patients with acute coronary syndromes. It is also used to prevent thrombosis in patients with non-cardiac surgery. 2. Clopidogrel is a prodrug that is converted to its active form by the enzyme CYP2C19.
	Medications on Admission: atenolol 25 mg daily aspirin 81 mg daily lipitor 10 mg QOD (every other am) prednisone 10 mg daily tamsulosin SR 0.4 mg evening multivitamin 1 tab daily fish oil capsule 1000 mg twice a day [s2] systane lubricant eye [e2] drops 1 gtt [s1] TID [e1].	The relation between “TID” and “systane lubricant eye” is “ Frequency-Drug ”.	I have been using nonpreserved systane eye drops for a few years.
3. Interpretable hallucination: the LLM generated a different way of the answer that is interpretable.	OXYCODONE - 20 mg Tablet Sustained Release 12 hr - 3 (Three) Tablet (s) by mouth every morning (60 mg), 1 tablet every 2 pm (20 mg) and 3 tablets every evening (60 mg) [s2] PANTOPRAZOLE [e2] [PROTONIX] - 40 mg Tablet , Delayed Release (E.C.) - [s1] 1 [e1] Tablet (s) by mouth once day	The relation between “1” and “PANTOPRAZOLE” is “ Dosage-Drug ”.	The relation type between the Drug entity “PANTOPRAZOLE” and Dosage entity “1” is “has_dosage”.
	Disp: * 30 Tablet (s) * Refills: * 2 * 3. [s2] Fluticasone - Salmeterol [e2] 250 - 50 mcg / Dose Disk with Device Sig: One (1) [s1] Disk with Device [e1] Inhalation Hospital 1 (2 times a day).	The relation between “Disk with Device” and “Fluticasone - Salmeterol” is “ Form-Drug ”	The relation type between the Drug entity “Fluticasone - Salmeterol” and Form entity “Disk with Device” is “is_a”.



1. Clinical Concept Extraction						
Datasets		2018 n2c2			2022 n2c2	
Model	Precision	Recall	F1	Precision	Recall	F1
BERT	0.8887	0.8728	0.8807	0.8160	0.8483	0.8318
GatorTron	0.8759	0.9038	0.8896	0.8181	0.8508	0.8341
BERT-MRC	0.9159	0.8942	0.9018	0.8496	0.8353	0.8424
GatorTron-MRC	0.9199	0.9012	0.9059	0.8521	0.8396	0.8451
GatorTronGPT-5B	0.9168	0.9057	0.9045	0.8598	0.8415	0.8589
GatorTronGPT-20B	0.9189	0.9081	0.9060	0.8615	0.8424	0.8615
2. Clinical Relation Extraction						
Datasets		2018n2c2			2022 n2c2	
Model	Precision	Recall	F1	Precision	Recall	F1
BERT	0.9598	0.9438	0.9545	0.7998	0.7620	0.7807
GatorTron	0.9719	0.9482	0.9599	0.7970	0.7631	0.7798
BERT-MRC	0.9722	0.9489	0.9604	0.8432	0.8371	0.8415
GatorTron-MRC	0.9724	0.9488	0.9605	0.8445	0.8390	0.8428
GatorTronGPT-5B	0.9734	0.9492	0.9610	0.8551	0.8472	0.8520
GatorTronGPT-20B	0.9746	0.9506	0.9614	0.8560	0.8491	0.8529
3. End-to-end Clinical Concept Extraction and Normalization						
SemEval-2015 task 14						
Dataset	Strict Precision	Strict Recall	Strict F1	Relax Precision	Relax Recall	Relax F1
Model						
ezDI (best in challenge)	0.783	0.732	0.757	0.815	0.761	0.787
ULisboa	0.779	0.705	0.740	0.806	0.729	0.765
UTH-CCB	0.778	0.696	0.735	0.797	0.714	0.753
GatorTronGPT-5B	0.810	0.751	0.782	0.833	0.769	0.795
GatorTronGPT-20B	0.812	0.772	0.791	0.839	0.785	0.813
4. Clinical Abbreviation Disambiguation						
Dataset	UMN abbreviation			5. Natural Language Inference		
Model	Precision	Recall	F1	Dataset	MedNLI Accuracy	
BiLSTM	0.8810	0.8960	0.8840	Model		
DistilBERT	0.9254	0.9358	0.9263	BioBERT	0.8050	
BioBERT	0.9479	0.9597	0.9505	ClinicalBERT	0.8270	
BlueBERT	0.9430	0.9518	0.9432	BioMegatron	0.8390	
GatorTronGPT-5B	0.9854	0.9832	0.9842	GatorTron	0.8670	
GatorTronGPT-20B	0.9849	0.9830	0.9836	GatorTronGPT-5B	0.8832	
				GatorTronGPT-20B	0.8946	
6. Medication Attribute Filling. Dataset: 2022 n2c2 Track 1						
	Event	Context				
Model	F1	F1				
GatorTron	0.9379	0.9126				
GatorTronS	0.9362	0.9080				
RoBERTa	0.8588	0.8417				
RoBERTa-MIMIC	0.9251	0.9121				
GatorTronGPT-5B	0.9332	0.8937				
GatorTronGPT-20B	0.9346	0.8941				
7. Progress Note Understanding						
Dataset		2022 n2c2				
Model		F1				
CMU (challenge best)		0.8212				
Yale		0.8133				
UWisc		0.8119				
UDuisburg		0.8034				
GatorTronGPT-5B		0.7875				
GatorTronGPT-20B		0.7954				



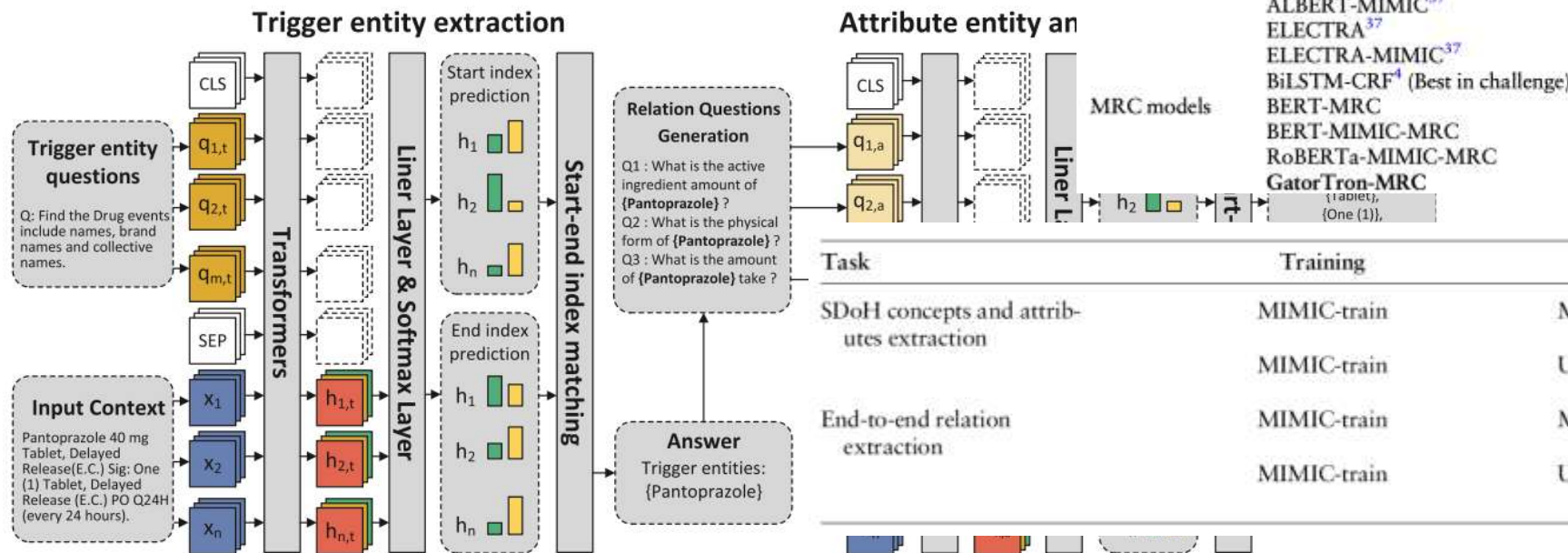
Cheng Peng, PhD

Prevail – Transfer learning

Research and Applications

Clinical concept and relation extraction using prompt-based machine reading comprehension

Cheng Peng ¹, Xi Yang ^{1,2}, Zehao Yu¹, Jiang Bian ^{1,2}, William R. Hogan ¹, and
Yonghui Wu ^{1,2,*}



		Strict scores			Lenient scores		
		Precision	Recall	F1	Precision	Recall	F1
Non-MRC models	LSTM-CRFs ⁸	0.8893	0.8728	0.8810	0.9392	0.9184	0.9287
	BERT ³⁷	0.8887	0.8728	0.8807	0.9398	0.9174	0.9285
	BERT-MIMIC ³⁷	0.8835	0.8871	0.8853	0.9367	0.9349	0.9353
	RoBERTa ³⁷	0.8821	0.8804	0.8812	0.9342	0.9276	0.9309
	RoBERTa-MIMIC ³⁷	0.8927	0.8888	0.8907	0.9425	0.9333	0.9379
	ALBERT ³⁷	0.8772	0.8766	0.8769	0.9311	0.9255	0.9283
	ALBERT-MIMIC ³⁷	0.8776	0.8909	0.8842	0.9272	0.9363	0.9317
	ELECTRA ³⁷	0.8689	0.8781	0.8735	0.9281	0.9314	0.9286
	ELECTRA-MIMIC ³⁷	0.8814	0.8857	0.8836	0.9319	0.9387	0.9325
	BiLSTM-CRF ⁴ (Best in challenge)	0.8973	0.8939	0.8956	0.9461	0.9376	0.9418
	BERT-MRC	0.9159	0.8942	0.9018	0.9651	0.9202	0.9440
	BERT-MIMIC-MRC	0.9187	0.8978	0.9031	0.9679	0.9221	0.9489
MRC models	RoBERTa-MIMIC-MRC	0.9165	0.8964	0.9020	0.9658	0.9210	0.9465
	GatorTron-MRC	0.9199	0.9012	0.9059	0.9708	0.9281	0.9506
Task	Training	Test		Model	Strict F1		
SDoH concepts and attributes extraction	MIMIC-train	MIMIC-test		GatorTron	0.8341		
				GatorTron-MRC	0.8452		
	MIMIC-train	UW-test		GatorTron	0.7488		
End-to-end relation extraction				GatorTron-MRC	0.8124		
	MIMIC-train	MIMIC-test		GatorTron	0.6395		
				GatorTron-MRC	0.7364		
	MIMIC-train	UW-test		GatorTron	0.5526		
				GatorTron-MRC	0.7122		

Prevail – Few-shot Zero-shot

Data and text mining

Advancing entity recognition in biomedicine via instruction tuning of large language models

Vipina K. Keloth ¹, Yan Hu ², Qianqian Xie ¹, Xueqing Peng ¹, Yan Wang ¹, Andrew Zheng ³, Melih Selek ⁴, Kalpana Raja ¹, Chih Hsuan Wei ⁵, Qiao Jin ⁵, Zhiyong Lu ⁵, Qingyu Chen ^{1,5,*}, Hua Xu^{1,*}

Model	Dataset				
		Strict/partial match	NCBI (disease) P/R/F1	BC5CDR (chemical) P/R/F1	BC2GM (gene) P/R/F1
GPT-3.5 (zero-shot)	Strict		0.483/0.286/0.359	0.743/0.448/0.559	0.302/0.430/0.355
	Partial		0.707/0.425/0.531	0.860/0.521/0.649	0.507/0.715/0.593
GPT-3.5 (five-shot)	Strict		0.515/0.230/0.318	0.235/0.721/0.355	0.450/0.351/0.395
	Partial		0.571/0.260/0.358	0.256/0.781/0.385	0.725/0.517/0.603
GPT-4 (zero-shot)	Strict		0.484/0.502/0.493	0.814/0.668/0.734	0.406/0.473/0.437
	Partial		0.665/0.723/0.692	0.899/0.749/0.818	0.689/0.752/0.719
GPT-4 (five-shot)	Strict		0.515/0.682/0.586	0.788/0.835/0.811	0.459/0.536/0.494
	Partial		0.589/0.789/0.675	0.837/0.882/0.859	0.755/0.791/0.772
PubMedBERT	Strict		0.868/0.877/0.873	0.924/0.942/0.933	0.845/0.816/0.830
	Partial		0.915/0.936/0.926	0.945/0.967/0.956	0.967/0.931/0.949
PMC-LLaMA	Strict		0.877/0.851/0.864	0.935/0.870/0.902	0.834/0.823/0.828
	Partial		0.938/0.914/0.926	0.957/0.892/0.924	0.959/0.943/0.951
BioNER-LLaMA1	Strict		0.869/0.882/0.876	0.935/0.916/0.926	0.841/0.837/0.839
	Partial		0.935/0.954/0.944	0.956/0.938/0.947	0.958/0.952/0.955
BioNER-LLaMA2	Strict		0.874/0.886/0.880	0.936/0.920/0.928	0.832/0.836/0.834
	Partial		0.941/0.957/0.949	0.957/0.944/0.951	0.955/0.956/0.956

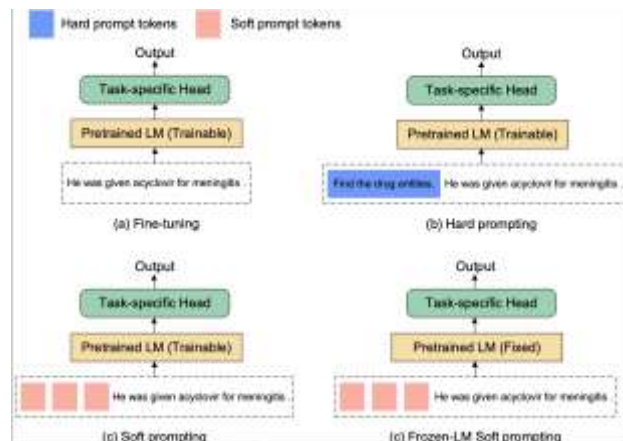
LLM Mirages

- Prompting solve all problems?
 - Limited control power – compared with fine-tuning
 - Not robust – works for some instances, don't work for other instances
 - Not controllable – no systematic way to develop good prompts
- LLMs are better than smaller NLP models?
 - For Artificial General Intelligence tasks
- Impact to classical clinical NLP tasks?
 - Clinical concept/relation extraction, text classification, world sense disambiguation, etc. – Moderate impact
 - You can optimize a small model to outperform LLMs
- Few-shot/zero-shot good enough for real-world applications?
 - Still a large gap to state-of-the-art many-shot learning models.

Prompt tuning – make machine learn robust, controllable prompts

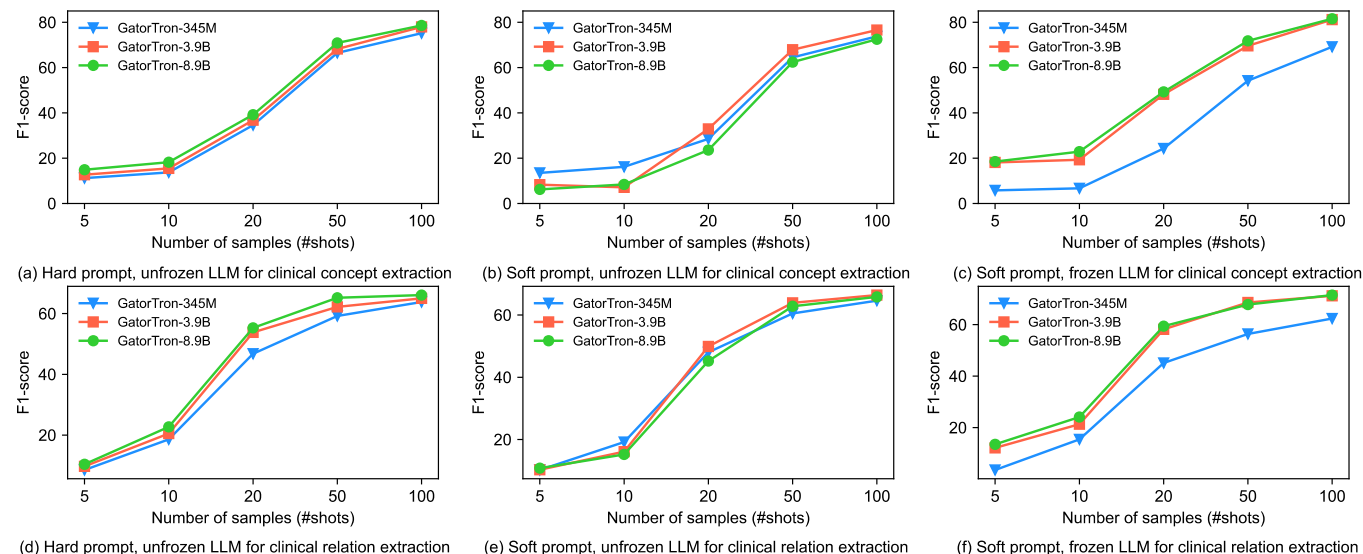


Cheng Peng, PhD



- Machines learn soft prompts better than humans
- Frozen LLMs has better few-shot learning ability and transfer learning ability
- Frozen LLMs require large models

Corpus	Models	Training strategies			
		No prompt; Unfrozen LLM	Hard prompt; Unfrozen LLM	Soft prompt; Unfrozen LLM	Soft prompt; Frozen LLM
2018 n2c2	BERT	0.8807	0.9018	0.9095	0.8598
	BERT-MIMIC	0.8853	0.9031	0.9080	0.8676
	RoBERTa	0.8812	0.9016	0.9076	0.8490
	RoBERTa-MIMIC	0.8907	0.9020	0.9081	0.8621
	GatorTron-345M	0.8870	0.9050	0.9112	0.8650



Challenge – Expensive



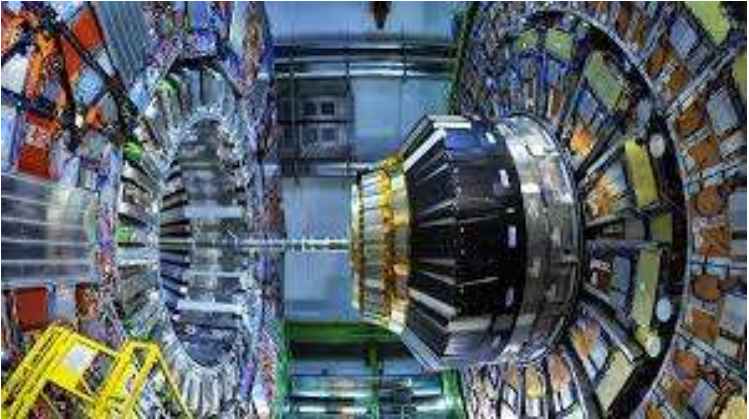
\$10 ~20 million



EVU machine – \$150 million



\$50 ~100 million

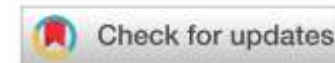


Large Hadron Collider – \$4.5 billion

Challenge – lack of clinical LLMs

REVIEW ARTICLE

OPEN



The shaky foundations of large language models and foundation models for electronic health records

Michael Wornow ¹✉, Yizhe Xu², Rahul Thapa², Birju Patel ², Ethan Steinberg ¹, Scott Fleming ², Michael A. Pfeffer^{2,3}, Jason Fries² and Nigam H. Shah ^{2,3,4,5}

The success of foundation models such as ChatGPT and AlphaFold has spurred significant interest in building similar models for electronic medical records (EMRs) to improve patient care and hospital operations. However, recent hype has obscured critical gaps in our understanding of these models' capabilities. In this narrative review, we examine 84 foundation models trained on non-imaging EMR data (i.e., clinical text and/or structured data) and create a taxonomy delineating their architectures, training data, and potential use cases. We find that most models are trained on small, narrowly-scoped clinical datasets (e.g., MIMIC-III) or broad, public biomedical corpora (e.g., PubMed) and are evaluated on tasks that do not provide meaningful insights on their usefulness to health systems. Considering these findings, we propose an improved evaluation framework for measuring the benefits of clinical foundation models that is more closely grounded to metrics that matter in healthcare.

npj Digital Medicine (2023)6:135; <https://doi.org/10.1038/s41746-023-00879-8>

Challenge - hallucination

Patient is a human female lives in Ylesia, who is a Glitterstim user frequently overdosed on the spice. After Glitterstim overdose, the patient appears light glazed off, unable to complete long-term memory tasks, but quick to react, to the others in trouble. Motivated to keep heroic till the end, while aiming to leave during lucid intervals.

Patient is a human female lives in Ylesia, who is a Glitterstim user frequently overdosed on the spice. After Glitterstim overdose, the patient appears light teal colored, slowly losing pigment to black. Notably her hair has turned white before becoming black to suggest a potential genetic mutation the mold colonies have selected to take over Ylesia's native population. First seen in Pipeline Flowr's Forest - at a heavily disturbed Boglands environment. The latter to create a reservoir of living subsurface hydroxides which can be then cycled to extracellular synthesis, and to promote plankton biomass productivity to cycle essential biochemicals for energy (taking advantage of internal destabilization) with The Pronomia behind it to keep it profitable. Lacking official sanction, Human harvesting in the long term will raise tensions with The Government to the point of world war and the discovery of such activity.



Cheng Peng, PhD

Research and Applications

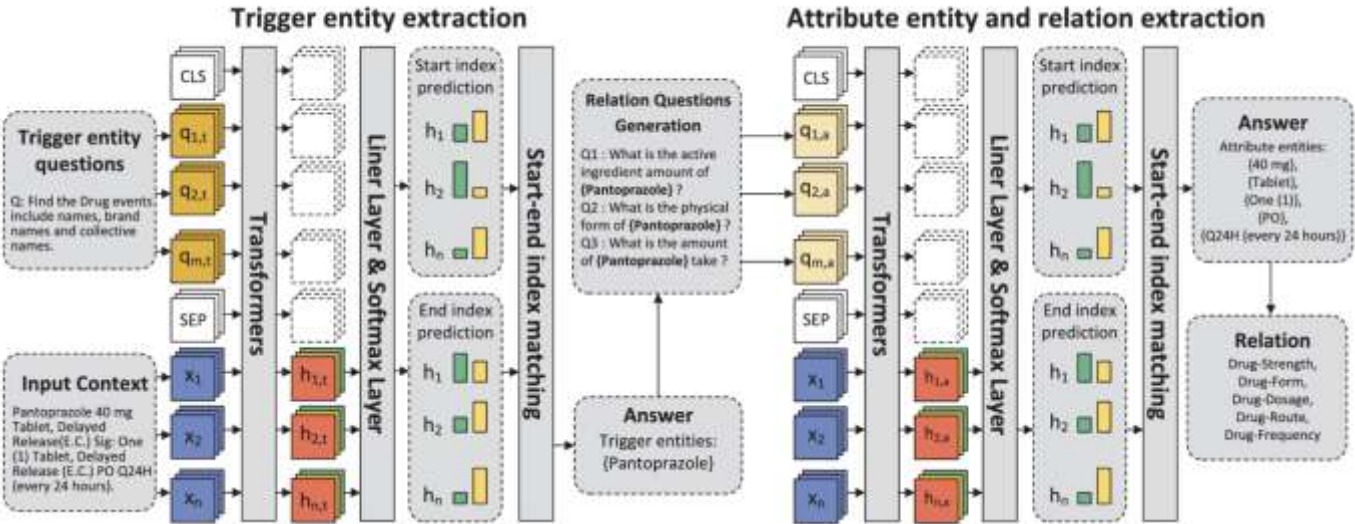
Clinical concept and relation extraction using prompt-based machine reading comprehension

Cheng Peng ¹, Xi Yang ^{1,2}, Zehao Yu ¹, Jiang Bian ^{1,2}, William R. Hogan ¹, and Yonghui Wu ^{1,2,*}

Hallucinations?

Drug	Find the drug events including names, brand names, and collective names.
Strength	Find the strength events that are the amount of drug in a given dosage.
Form	Find the form events that are the physical form of given drug or medication.
Dosage	Find the dosage events that are the amount of a medication used in each administration.
Frequency	Find the frequency events that indicate how often each dose of the medication should be taken.
Route	Find the route events that are the path by which a drug is taken into the body.
Duration	Find the duration events that indicate how long the medication is to be administered.
Reason	Find the reason events that are the medical reason for which the medication is given.
ADE	Find the ADE events that are injuries resulting from a medical intervention related to drugs.

Task	Training	Test	Model	Strict F1
SDoH concepts and attributes extraction	MIMIC-train	MIMIC-test	GatorTron	0.8341
			GatorTron-MRC	0.8452
	MIMIC-train	UW-test	GatorTron	0.7488
			GatorTron-MRC	0.8124
End-to-end relation extraction	MIMIC-train	MIMIC-test	GatorTron	0.6395
			GatorTron-MRC	0.7364
	MIMIC-train	UW-test	GatorTron	0.5526
			GatorTron-MRC	0.7122



Relation Questions	
Strength	What is the active ingredient amount of “Pantoprazole”?
Form	What is the physical form of “Pantoprazole”?
Dosage	What is the amount of “Pantoprazole” taken?
Frequency	How often each dose of “Pantoprazole” should be taken?
Route	What is the path of “Pantoprazole” taken into the body?
Duration	How long to take “Pantoprazole”?
Reason	What is the medical reason for giving “Pantoprazole”?
ADE	What are the injuries resulting from the use of “Pantoprazole”?

Challenges - Security and privacy

FORBES > BUSINESS

BREAKING

Samsung Bans ChatGPT Among Employees After Sensitive Code Leak

Siladitya Ray Forbes Staff

Covering breaking news and tech policy stories at Forbes.

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May 2, 2023, 07:17am EDT



Who come to see Dr.
Bob yesterday?

Where does he live?

Future directions

- Lower the computing cost
- Make prompt-based learning controllable, robust, and efficient
- Few-shot learning to be comparable to state-of-the-art
- Synthetic clinical text generation – can we develop NLP systems using pure synthetic clinical text? (Google's AlphaZero)

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 - **PCORI ME-2018C3-14754 (PI, contact)** - NLP to Connect Social Determinants and Clinical Factors for Outcomes Research.
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 - **PCORI RI-FLORIDA-01-PS1 (Director of Natural Language Processing)** - OneFlorida+ Phase 3 Clinical Research Network.
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 - NIH/NIA R01AG080991 - Eligibility criteria design for Alzheimer's trials with real-world data and explainable AI.
 - NIH/NIA R01AG084178 - Post-Acute Sequelae of SARS-CoV-2 Infection and Subsequent Disease Progression in Individuals with ADRD.
 - NIH/NIA R01AG084236 - ACTS (AD Clinical Trial Simulation): Developing Advanced Alzheimer's Disease Clinical Trial Simulation System.
 - NIH/NIA R21AG068717 - Optimizing the Population Representativeness of Older Adults in ADRD Clinical Trials.
 - NIH/NIA R21/R33 R33AG062884 - Advancing Interdisciplinary Science of Aging through Identification of Iatrogenic Complications (UF-ECLIPSE).
 - NIH/NIAID R01AI172875 - Artificial Intelligence and counterfactually actionable response to end HIV (AI-CARE HIV).
 - NIH/NHLBI R01HL169277 - Identifying pediatric asthma subtypes using novel privacy-preserving federated machine learning methods.
 - NIH/NIMH R01MH121907 - Linking VA and non-VA data to study the risk of suicide in chronic pain patients.
 - NIH/NCI R01CA284646 - Advancing Precision Lung Cancer Surveillance and Outcomes in Diverse Populations (PLuS2).
 - NIH/NCI R37CA272473 - De-implementation of inappropriate thyroid ultrasound.
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UF GPU SUPERPOD

Featuring NVIDIA DGX A100 640G

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